

Final Year Project

Automated Recognition of Tactics in Chess: Exploratory Analysis of the Role of Tactics in Gameplay

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Abstract

This project focuses on automating the identification of tactics, also commonly referred to as *tactical motifs*, in chess games by leveraging advanced chess engines and external domain knowledge. A tactical motif in chess refers to a specific pattern or combination of moves aimed at gaining a tactical advantage, such as a material advantage or checkmate threat. By analysing millions of games from the LiChess.org open database, this project aims to offer unique insights into the game's strategic intricacies. The development of *Tactical Recognition Algorithms (TRAs)* will enable players to assess their tactical capabilities and identify areas for improvement. Through this endeavour, the project seeks to contribute to the ongoing exploration of chess mastery and provide valuable resources for players seeking to enhance their skills.

Chapter 1: Introduction

Chess is a highly theoretical and tactical game that has been studied and played for a long time. With the introduction of computers and artificial intelligence (AI) began an exploration into their capabilities and utilities in the realm of chess. Today, there are superhuman AI chess engines that can consistently and convincingly beat the best Grandmasters, however, their non-human approach to the game makes them incompatible learning tools when used in isolation.

In the realm of chess training resources, players have access to a plethora of high-quality materials, ranging from free online platforms to subscription-based services. A common form of tactical training is through puzzles, which present players with tactical problems that have a single, clear solution. However, puzzles have their limitations; players approach them with the foreknowledge that a solution exists, unlike in actual games where the tactical landscape is often less clear-cut. Notably, Chess.com, a leading all-in-one chess website, recently introduced a groundbreaking tool for players to analyse tactics played in their games. Nevertheless, this resource is subscription-based and analyses a limited selection of tactics. Despite this development, there remains a clear gap in the market for a freely accessible resource that allows players to analyse their tactical play within the context of their games on an ongoing basis.

Tactics hold a profound significance in chess gameplay, a sentiment famously captured by *Richard Teichmann's* quote, "*Chess is 99% tactics*". While this assertion lacks empirical evidence, it has sparked enduring debate within the chess community. Nonetheless, it is widely acknowledged that tactics play a pivotal role in the game, influencing the outcome of encounters at all skill levels.

This project aims to leverage state-of-the-art chess engines and external domain knowledge to automate the identification of tactical positions in chess games. To achieve this, *Tactical Recognition Algorithms (TRAs)* are designed and implemented, enabling the extraction of tactics from thousands of games from the LiChess database. By outlining questions concerning the role of tactics in chess and scrutinising the extracted tactics for answers, this project aims to offer unique insights into the game's strategic intricacies. Motivated by my intrigue in chess and a desire to contribute to the chess community, this project seeks to offer novel insights into the age-old debate of how mastery is reached in chess.

Chapter 2: Related Work and Ideas

The fusion of artificial intelligence (AI) and chess has sparked a captivating journey, from early human-computer competitions to the era of superhuman chess engines. While these engines excel on the board, their moves often defy human comprehension, posing a challenge in leveraging AI for player improvement. This review explores the intersection of human expertise with cutting-edge AI methodologies in chess research, aiming to shed light on strategies to enhance player performance and bridge the gap between human and machine intelligence.

2.1 Chess and AI Research

Chess has played a pivotal role in the advancement of artificial intelligence (AI) research, serving as a fertile ground for exploring complex reasoning problems within a relatively simple domain [1]. The game's well-defined rules, finite board, and built-in performance criteria make it an ideal testbed for developing and evaluating AI algorithms [1]. One of the defining characteristics of chess is its rich complexity, which arises from the vast number of possible board configurations and the intricate interplay of strategic and tactical elements. This complexity presents a formidable challenge for AI researchers, requiring the development of algorithms capable of exploring the game tree, evaluating positions, and making strategic decisions.

Chess and *Drosophila*, the common fruit fly, share narrow goals and limitations as experimental technologies in AI and cognitive science. The phrase "*chess is the Drosophila of AI*" underscores the importance of chess as a model system for studying intelligence and cognition [2]. Researchers have turned to chess to investigate key aspects of intelligent behaviour, such as problem-solving, pattern recognition, and decision-making.

The historical significance of chess in AI research is exemplified by milestones such as Deep Blue and AlphaZero. In 1997, IBM's Deep Blue made headlines by defeating world chess champion Garry Kasparov in a six-game match, marking the first time a computer had defeated a reigning world champion under standard chess tournament conditions [3]. Deep Blue's victory demonstrated the potential of AI to rival human expertise in complex cognitive tasks and sparked widespread interest in the field [2].

More recently, DeepMind's AlphaZero made waves in 2017 by achieving superhuman performance in not only chess but also the games of Go and Shogi [4]. AlphaZero's achievement was particularly noteworthy because it learned to play solely through reinforcement learning. This approach represented a departure from traditional chess programming methods, which relied on human-refined evaluation functions, and showcased the power of machine learning techniques in advancing AI research [5].

In contrast to AlphaZero's approach, traditional chess engines like Stockfish utilise brute-force search algorithms combined with sophisticated evaluation functions. Stockfish, the current world champion in computer chess, employs alpha-beta pruning and other optimisation techniques to evaluate millions of positions per second. Alpha-beta pruning is an adjustment to minimax search which improves evaluation by discarding move variations which will never be reached in optimal play [6]. The accomplishments of Stockfish surpass those of any other chess engine, having won more than

10 Top Chess Engine Championships and dominating the Computer Chess Championship since 2018, winning the first six events and more [7]. Stockfish's prowess is evident in its widespread adoption by popular chess websites such as LiChess and Chess.com for their analysis tools, showcasing its recognition and utilisation within the chess community [8][9].

Additionally, Stockfish is free and open-source, first released in November 2008 [10]. Developed by an entire community of contributors, including enthusiasts and developers, Stockfish continues to evolve and improve [11][12]. In September 2020, Stockfish 12 was released, marking a significant milestone as it absorbed the Stockfish+NNUE project (NNUE stands for Efficiently Updatable Neural Network) [8]. This move enhances Stockfish's capabilities by integrating the evaluation abilities of a neural network engine with its raw computational power, creating a formidable combination of traditional and neural network-based approaches.

2.2 Aligning Superhuman AI with Human Behaviour

Chess computers have served as training partners for humans for some time. However, they are not perfect substitutes for human players as they don't expose players to human mistakes [13]. In recent research, there has been a concerted effort to bridge the gap between human and AI methodologies in chess. The aim is to enhance player performance and create AI systems that exhibit behaviour more akin to human decision-making. Two significant studies have delved into this endeavour.

Das & Chernova investigated the efficacy of providing explanations for AI moves in improving player performance in chess. They developed an algorithm capable of generating human-interpretable rationales for AI moves, termed the Rationale-Generating Algorithm (RGA) [14]. When players utilised these explanations, they experienced a notable improvement in their performance. The algorithm works by analysing the current game state and providing explanations for the AI's move choices based on factors such as piece activity and strategic considerations. The RGA aims to enhance player understanding and decision-making by providing transparent insights into the AI's thought process. Fig. 2.1 illustrates the RGA in action, with explanations provided alongside AI moves.



Figure 2.1: Example of RGA [14]

McIlroy-Young et al. customised the AlphaZero engine to develop a model named Maia. Unlike traditional deep reinforcement learning methods, which struggle to accurately model human behaviour, Maia incorporates additional features and heuristics inspired by human gameplay [15].

By analysing human games, Maia demonstrates superior alignment with human decision-making compared to existing chess engines. It employs a combination of machine learning techniques, including neural networks and statistical analysis, to learn patterns and strategies from human games. Additionally, Maia's insights into human playstyles open up possibilities for predicting human errors and extending the application of these techniques beyond chess to other domains where human-like decision-making is crucial.

Both studies employed sophisticated computational methodologies to enhance player performance and imbue AI systems with human-like characteristics in the chess domain. While the RGA study prioritised the development of explanatory AI systems, the Maia study focused on replicating human playstyles. The findings of both highlight the efficacy of their approaches in improving player performance. However, while RGA emphasised the importance of providing understandable explanations, Maia focused on achieving alignment with human behaviour. Both studies advocate for further exploration of explanatory AI systems to enhance player performance and extend the application of these techniques to domains beyond chess.

2.3 Theories of Chess Skill

In the quest to unravel the intricacies of chess skill, there are two prominent theories: the *recognition-association* theory and the *forward-search (SEEK)* theory. These theories help us grasp how experts develop skills and make decisions in chess.

The recognition-association theory, proposed by *Chase and Simon* (1973), posits that chess expertise is rooted in the ability to recognise and recall specific patterns of pieces on the board [16]. According to this theory, chess masters excel because they quickly recognise and remember specific patterns on the board. They don't foresee many moves ahead but rely on a wide range of memorised patterns to guide their decisions. In contrast, the "*commonsense*" forward-search theory known as the *SEEK model*, outlined by *Holding* (1985) [17], says that chess expertise comes from efficient search processes and accurate judgment of end positions. Unlike the recognition-association theory, which focuses on pattern recognition, SEEK emphasises analytical thinking and strategic planning.

The interplay between slow, analytical processes (SEEK) and fast, intuitive heuristics (recognition-association) is crucial in chess. Chess is known for its complexity, requiring both types of thinking [18][16]. The Elo rating system, which is used to measure players' skill level [19], reflects this complexity by considering separate ratings for each game variant, which differ in time constraints, and consequently thinking speeds [20].

There's ongoing debate regarding the relative importance of slow search versus fast pattern recognition in chess performance. Some argue that fast pattern recognition is crucial, citing better move quality in rapid games and accurate prediction of opponents' moves [21][22]. Others suggest that time constraints in fast games impair players' ability to think through their moves, affecting performance [23][24]. However, while the recognition-association theory has amassed a lot of indirect evidence, it has been shown that higher-rated players search more extensively and deeply, with results suggestive that the ability to evaluate positions reached in search differs only in skill level [25].

Recent studies challenge the dichotomy between slow and fast processes, emphasising their interdependence and adaptive nature [26][27][23]. Expert players demonstrate a remarkable ability to switch between the two, using their knowledge and experience to make quick, effective decisions [28][29]. Longitudinal studies provide additional evidence suggesting that chess players at the highest expert level displayed a greater consistency regarding the interrelationships of both processes,

which can lead to more efficient decision-making strategies [30].

In conclusion, theories of chess skill offer valuable insights into how experts think and perform in the game. While there is no research specifically measuring the ability of players to identify and execute tactical motifs, this ability and the recognition-association theory are intrinsically linked. Tactical motifs are one of the many branches of chess knowledge, which is used in the evaluation of positions in forward-search [17][25]. With this ability closely tied to both theories, tactical proficiency is likely correlated to chess skill.

2.4 The roles of practice and intelligence in Chess

Since the inception of chess, there has been deliberation over what factors contribute to the differences in skill across all playing levels. The seminal work of *De Groot* in understanding perception in chess, on which the recognition-association theory was founded, formed the notion that chess mastery was acquired in the long-term memory acquisition of positional patterns, thus supporting the notion that achieving mastery in the sport took nothing but thousands of hours of practice [31][16].

For well over the past century in psychology, one of the most hotly debated questions is the relationship between cognitive ability and expertise [32]. Chess is the original domain for research in expertise, and it is an ideal domain for much the same reasons why it has been fundamental in AI research: it is one of the single most studied domains in research on expertise, it has an objective measure of skill, and it is a complex and purely intellectual sport. Given this, it is surprising that inconsistent evidence supports this question [32].

As discussed in the previous section, there has been plenty of debate over what factors contribute to chess expertise. Not too long ago, some claimed that "*there is no correlation between IQ and expert performance in chess*" [33]. More recent work has found significant correlations between chess skill and cognitive ability [34][35]. The monumental work of *Burgoyne et al.* in which they conducted the first-ever meta-analysis of this question revealed results suggestive that cognitive ability contributes meaningfully to individual differences in chess skill [32]. An interesting finding was that chess skill correlated more strongly with numerical ability than with verbal ability or visuospatial ability. This perhaps supports the aforementioned SEEK model, where search evaluation is the defining factor of chess skill.

The roles of practice and motivation in chess have long been studied too. The seminal work of *Ericsson*, in which he outlined the deliberate practice framework, provided evidence supporting the argument that deliberate practice plays a critical role in the acquisition of chess expertise [36][37]. Later work indicated limitations in *Ericsson's* theory, highlighting the differences in skill between masters and experts who had done similar amounts of deliberate practice. Suggested reasons for differences in chess ability included variations in age of initial exposure and motivation levels [38]. Despite these limitations, several studies have found a significant, positive correlation between practice and chess skill [39][36]. *Gobet & Campitelli (2011)* conclude that abundant deliberate practice is required to reach chess mastery [40].

An interesting study of recent times was designed to tackle the common pitfalls of previous research and to study the contribution of practice, intelligence, and motivation in young children who had just begun to learn chess. The results suggest that general intelligence and motivation influence chess performance at the early stages of development [38].

In conclusion, the journey to mastery in chess intertwines the pivotal roles of practice, cognitive

abilities, and motivation. While extensive practice is essential for skill acquisition, early developmental factors set the stage for one's ultimate ceiling of mastery. This nuanced understanding underscores the complexity of expertise development in chess and highlights the need for comprehensive research approaches to unravel its intricacies.

2.5 Tactical motif training and insights

The predominant method for training tactical motifs comes in the form of *puzzles*. Puzzles are tactical problems which only have a single, crystal-clear solution [41], thus solving them requires calculation and tactical fluidity. While most major chess-playing websites have a puzzle training feature, LiChess' one is particularly impressive, especially as it is free, unlike their competitors such as Chess.com. Every LiChess puzzle is tagged with one or several tactical "*themes*", and what differentiates LiChess from its competitors is its *Puzzle Dashboard*, which allows players to track their puzzle proficiency over time. Additionally, there are insights broken down by theme, with LiChess suggesting areas needing improvement. These are seen in Fig. 2.2.

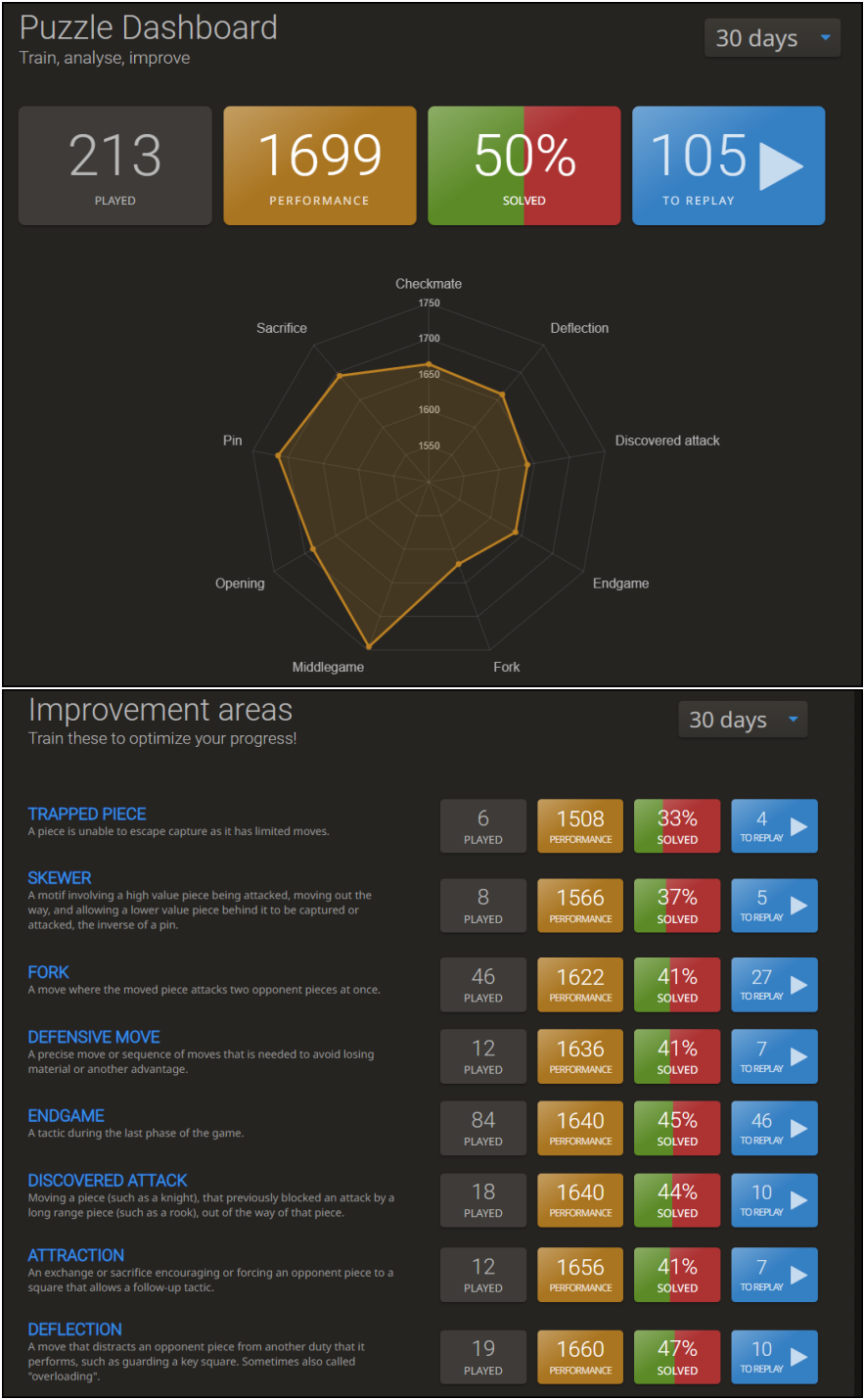
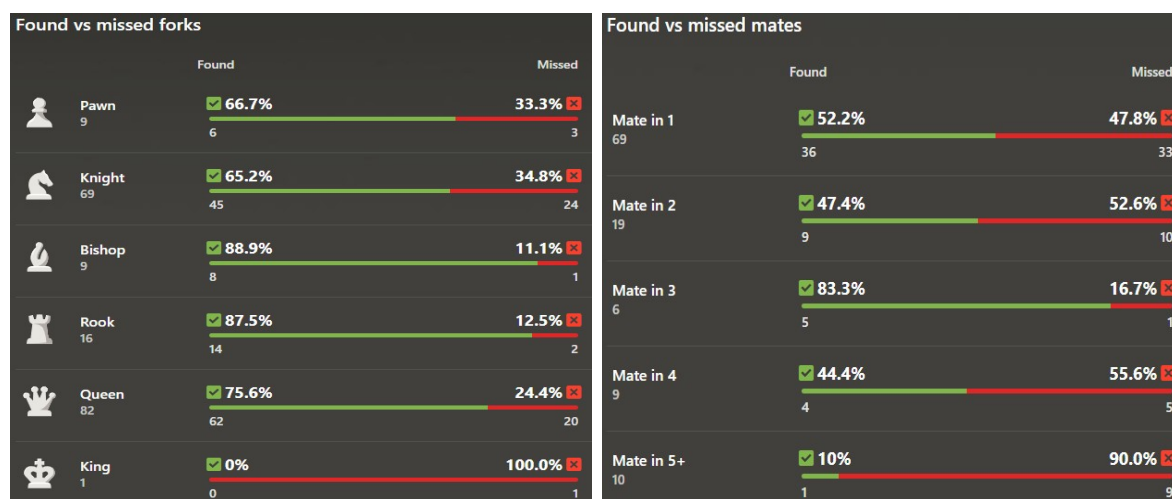


Figure 2.2: LiChess Puzzles: Dashboard (top), Improvement Areas (bottom) [42][43][44]

The puzzles on LiChess are extracted from games played on the website using Stockfish [41]. There are several themes which are automatically assigned to puzzles, although players suggest the tactical motif themes through the puzzle rating system. In this system, once players finish a puzzle they are prompted with several themes in which they assign a "thumbs up" or "thumbs down" according to whether each one is relevant to the puzzle or not. These ratings are then used to decide the themes assigned to the puzzles [41].

Chess.com have recently released a novel feature "*Insights*", which provides a thorough analysis of your games played on the website, including some insights for tactical motifs [45]. However, it is a "premium" feature, where use of it will require a paid subscription. The feature visualises the player's proficiency and use of three tactical motifs: Pins, Forks, and Checkmates. This can be seen in Fig. 2.3.

A found tactic in this instance is when the player played the top engine line and it involved one of the tactical motifs. Therefore, a missed tactic is when the player did not play the top engine line and it involved one of the tactical motifs. What is unknown is how exactly they are identifying these tactical motifs, although I presume they are using algorithms as opposed to a rating system like LiChess.



(a) Forks: Found rate by piece

(b) Checkmates: Found rate by piece

Figure 2.3: "Insights" feature on Chess.com [45]

The insights also include useful comparisons to similarly-rated players and prior performance. In Fig. 2.4a, you can see my performance in finding Pins with the three eligible pieces: bishop, rook, and queen. Similarly, you can see my past performance in Fig. 2.4b.

Evidently, there is a market for providing training and insights for tactical motifs, with Chess.com currently ahead of the curve. Although their current version of the feature provides insights into many more aspects of chess than just tactical motifs, their selection of tactical motifs is small. There is a gap in the market for a tool which continuously monitors a player's games for a wide variety of tactical motifs, providing insights into tactical play and comparing proficiency across varying Elos.

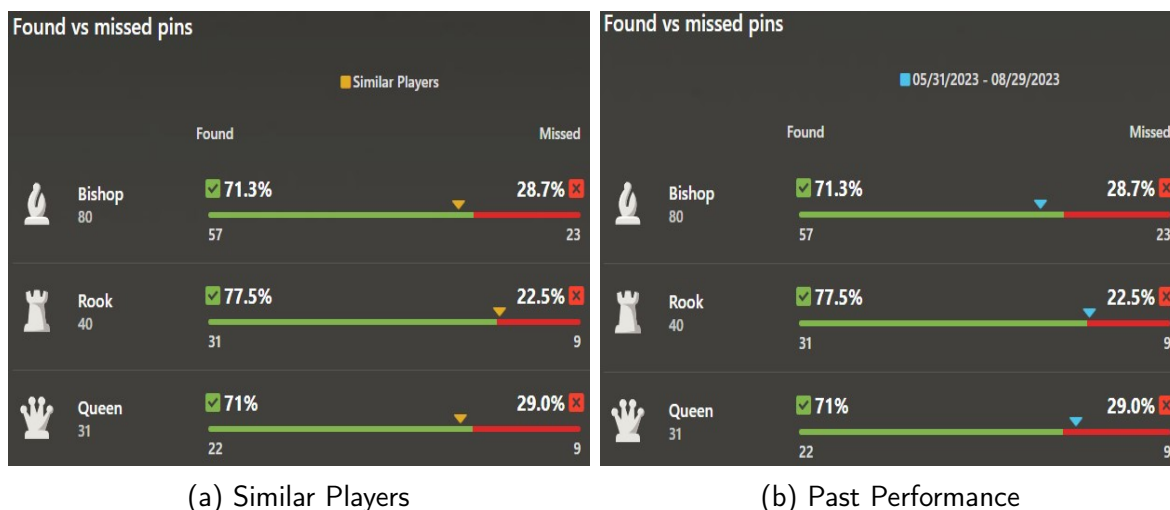


Figure 2.4: Pins Insights (from Chess.com) [45]

2.6 Conclusion

In this review of AI and chess, we have covered the rich tapestry of research and ideas shaping the intersection of human expertise and machine intelligence in the realm of chess. From the historical significance of chess in AI research to recent advancements such as AlphaZero, the journey has been marked by several milestones and discoveries.

We explored the dichotomy between human and AI decision-making in chess, highlighting efforts to imbue AI systems with human-like characteristics and enhance player performance. Theories of chess skill provided valuable insights into the cognitive mechanisms underlying expertise development, underscoring the importance of both analytical thinking and pattern recognition in chess mastery.

Furthermore, we delved into the pivotal roles of practice, intelligence, and motivation in chess, unravelling the complex interplay of factors that shape players' journey to mastery. From the foundational work of *De Groot* to recent meta-analyses on the relationship between cognitive ability and chess skill, we gained a nuanced understanding of the multifaceted nature of expertise development in chess.

As we conclude this review, it becomes clear that understanding the complexities of chess skill is an ongoing endeavour. By bridging the gap between human and AI methodologies, we strive to empower players with insights and tools to navigate the complexities of the chessboard and push the boundaries of human achievement in the age of artificial intelligence.

Chapter 3: Data Considerations

The data used in this project is from the open database provided by LiChess, which is released under the Creative Commons CC0 licence, which permits the use of the database for research purposes, among others [46]. It contains over 5 billion rated games, played on the website, in PGN format.

PGN (short for Portable Game Notation) is the standard for recording a game in a text file that is processible by computers [47]. The PGN typically records the entire sequence of moves in a chess game. It also stores additional information relevant to the game, such as the names of the players, the venue, the time control, the players' ratings, the game's results, and so on. Fig. 3.1 showcases an example of a PGN.

```
[Event "Live Chess"]  
[Site "Chess.com"]  
[Date "2020.03.25"]  
[Round "-"]  
[White "pdrpnht"]  
[Black "ColinStapczynski"]  
[Result "0-1"]  
[WhiteElo "1543"]  
[BlackElo "2241"]  
[TimeControl "180"]  
[Termination "ColinStapczynski won by resignation"]  
  
1. e4 d5 2. exd5 Qxd5 3. Nc3 Qa5 4. d4 c6 5. Nf3 Bf5 6. Bd3 Bxd3 7. Qxd3 e6 8.  
O-O Nf6 9. Bg5 Nbd7 10. Ne5 Qc7 11. Ne2 Nxe5 12. dxe5 Qxe5 13. Bxf6 gxf6 14.  
Rfe1 Bd6 15. Ng3 Qd5 16. Rad1 Qxd3 17. Rxd3 O-O-O 18. Red1 Be7 19. Ne4 Rxd3 20.  
Rxd3 Rd8 21. Rh3 f5 0-1
```

Figure 3.1: Example of a PGN [47]

PGNs are important as they allow players to replicate games with ease using chess software. Additionally, they can be used to analyse games using powerful chess engines such as Stockfish, which I will be using in this project. In Section 4.3, I outline the exact steps which were required to prepare the database for my analysis of the tactical elements of chess.

Chapter 4: Methods

The primary objective of this project is to unravel novel insights into the realm of tactical play in chess. With a plethora of areas to explore and insights to extract, I set out to answer the following questions:

- How often do tactics appear in gameplay?
- Does tactical proficiency correlate with skill?
- What tactics should players focus on?
- Does time affect tactical proficiency?

In this chapter, I outline the sequence of steps required to conduct this analysis, detail how I achieved each one, and provide reasoning for my approach. A high-level overview of the outline of methods is seen in Fig. [4.1](#)



Figure 4.1: Outline of Methods

4.1 Environment Setup

The project was completed using Jupyter Notebooks as the primary platform for code development, data preparation, and analysis. Jupyter Notebooks provide an interactive computing environment that seamlessly integrates code, visualisation, and explanatory text, facilitating a coherent and reproducible research workflow. Stockfish, a powerful open-source chess engine, which was crowned champion of the latest season of the Top Chess Engine Championship [7], was utilised for evaluating chess positions and generating optimal sequences of moves. Its ability to discover superior moves with remarkable speed makes it an ideal choice for analysing complex positions in just a few seconds [12]. The Python programming language serves as the backbone for implementing the *Tactical Recognition Algorithms (TRAs)* and interacting with the chess engine. Specifically, the Python-Chess package handled chess board representation, move generation, and position evaluation, enabling seamless integration with Stockfish for tactical analysis [48].

4.2 Tactical Recognition Algorithms

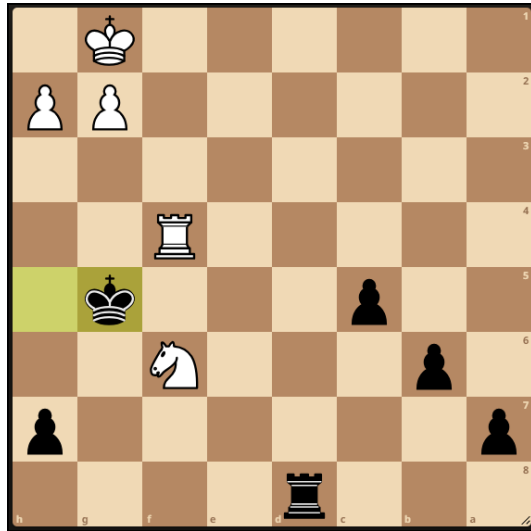
As there are several unique tactical motifs, often referred to as tactics, each with a set of defining parameters, it was necessary to design and implement one TRA for each tactic. While there is no concrete answer as to which tactics are the most important or the highest occurring, several high-level chess masters offer online courses covering the details and intricacies of a range of hand-picked tactics, which they believe are worth learning. One such course titled "*Tactics Masterclass*" by International Master *Levy Rozman* provides in-depth explanations of several tactics [49]. Below is a list of these tactics which I implemented into TRAs:

- Forks
- Pins
- Discovered Checks and Double Checks
- Skewer
- Undermine (Remove the Defender)

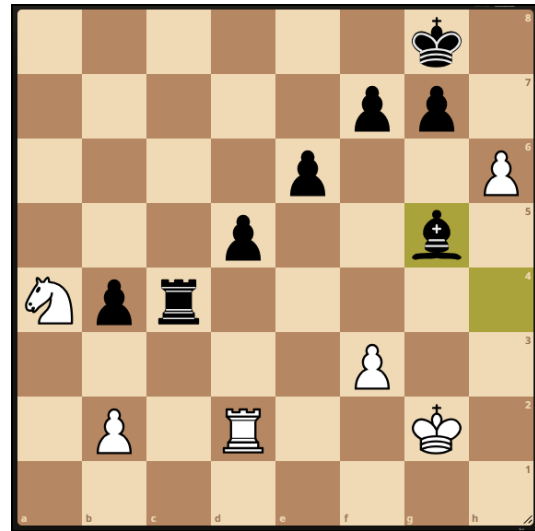
Next, I will describe each tactic and discuss the design and implementation of its TRA.

4.2.1 Forks

A Fork is a tactic whereby a single piece attacks two or more pieces at the same time. The attacking piece is called the forking piece; the pieces attacked are said to be forked [50]. Below are two examples of a Fork:



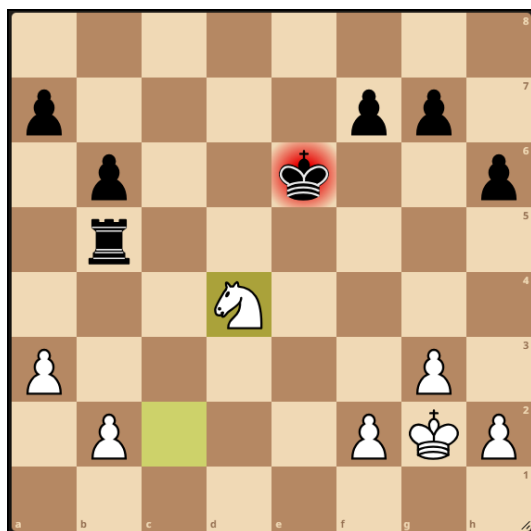
(a) King Fork



(b) Bishop Fork

Figure 4.2: Examples of Forks

Although every piece can deliver a Fork, the most striking Forks come from a knight. Because of the unique way it moves, a forking knight cannot be captured by the pieces it is threatening. The deadliest Forks are the ones that are forcing. Usually, those Forks involve an attack on the enemy king and create an inescapable threat [50]. Below are examples of each:



(a) A Forcing Fork



(b) Knight Fork

Figure 4.3: More Examples of Forks

Before I discuss the implementation of this TRA, it is important to understand the relative value of chess pieces. A relative value (or point value) is a standard value conventionally assigned to each piece. Piece valuations have no role in the rules of chess but are useful as an aid to assessing

a position [51]. The best-known point value system is depicted in Fig. 4.4, in which it assigns 1 point to a pawn, 3 points to a knight or bishop, 5 points to a rook and 9 points to a queen. However, it is important to note that valuation systems provide only a rough guide and the true value of a piece is very position-dependent. Machine learning has been employed to determine the true value of chess pieces, with the results indicating that, relative to the knight, the bishop and pawn are undervalued while the rook is overvalued [52]. The traditional system was employed for the implementation of the TRAs.

Symbol					
Piece	pawn	knight	bishop	rook	queen
Value	1	3	3	5	9

Figure 4.4: The Relative Value of Pieces [51]

Let's dive into the intricacies of the Fork TRA, given in Alg. 1, with a focus on the criteria which qualify and disqualify a move from being a Fork. The first criterion which eliminates a move from being a Fork is if the square that the move lands on is defended, a square is defended when there is at least one enemy piece which can legally move to it. To check this, I've written a function Alg. 2, which returns the number of enemy pieces which legally attack a given square. Additionally, for a move to qualify as a Fork, it must fulfil at least one of the two following criteria:

- Attack two or more undefended pieces
- Attack two or more pieces of higher relative value

To achieve this, we iterate over the attacked squares and tally the occurrences of each of the criteria.

4.2.2 Pins

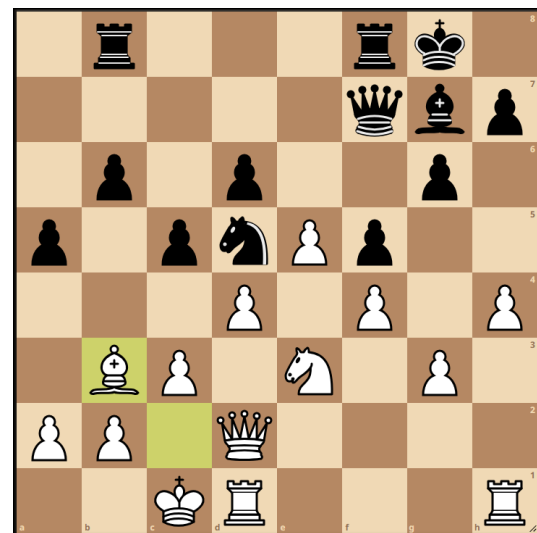
A Pin is a tactic whereby a ranged piece restricts one or more of your opponent's pieces. An Absolute Pin is the most potent version of this tactic, occurring when a piece is covering an attack on the king. Since it is illegal to make a move that would put your king in check, the pinned piece literally cannot move. Another version of this tactic is a Relative Pin, this happens when a piece is covering an attack on a higher-value piece (which is not the king). Although moving the pinned piece is not illegal, it may not be desirable [53]. Examples of both are depicted in Fig. 4.5.

For a move to be considered a Pin, regardless of it being absolute or relative, it must fulfil the following criteria:

- The piece moved is not under attack
- The piece attacked is not the king or a pawn
- It is a new Pin



(a) Absolute Pin



(b) Relative Pin

Figure 4.5: Examples of Pins

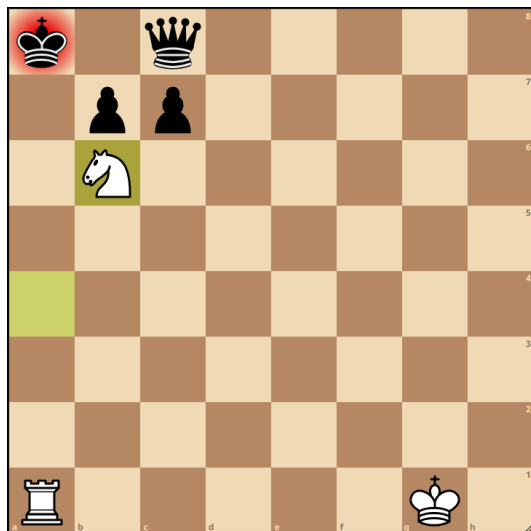
It is worth noting that a Pin can also be revealed, rather than directly created by the moving piece. High-level descriptions of the two TRAs are given in Alg. 3, and Alg. 4. The Absolute Pin TRA utilised the built-in method to check for Absolute Pins. The Relative Pin TRA required additional functionality, which is detailed later, as there were no built-in methods for this version of the Pin. The lengthy conditional ensures that relative-value logic is sound before confirming an instance of this Pin.

The additional functionality enables searching for an x-rayed piece and checking if a Relative Pin was already present, and are given in Alg. 10 and Alg. 11 respectively. An x-rayed piece is a piece which is directly behind a piece attacked by a ranged piece. For example, in Fig 4.5b the black queen is x-rayed by the white bishop.

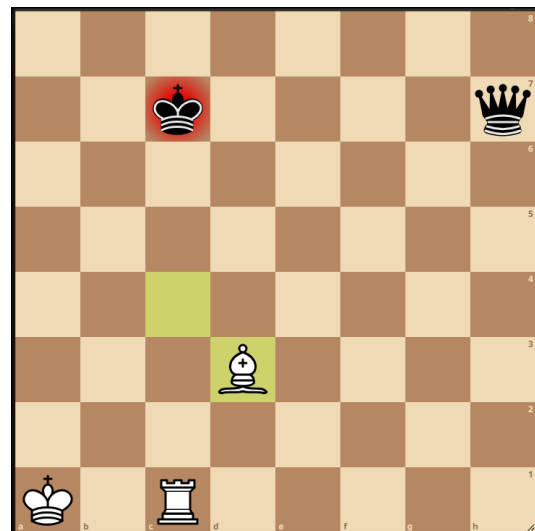
4.2.3 Discovered Checks and Double Checks

Discovered Checks and Double Checks fall under the same family of checking tactical motifs, as the names suggest, and both are quite straightforward. A Discovered Check occurs when a piece moves and it consequently reveals an attack on the opponent's king from another piece, resulting in a check [54]. This can be a particularly potent tactic, especially when the moving piece captures another piece or attacks a higher-valued piece. For example, in Fig. 4.6b the bishop directly attacks the opponent's queen while the rook puts the king in check, resulting in a devastating capture.

A Double Check is as obvious as the name suggests; two (or more) pieces put the opponent king in check simultaneously [55]. This tactic is always an instance of the Discovered Check but with at least one additional piece putting the opponent king in check. You can see an instance of this in Fig. 4.6a where the knight and rook are simultaneously checking the king, resulting in a forced checkmate on the following move.



(a) Double Check



(b) Discovered Check

Figure 4.6: Examples of Double Checks and Discovered Checks

The high-level descriptions of the TRAs are given in Alg. 5 and Alg. 6, the simplicity of these tactics is captured by these straightforward algorithms. Built-in methods of the Python-Chess package for checking if a board is in check, counting the number of attackers of a square, and replacing pieces on a board helped shorten the implementation. In particular, the Discovered Check TRA leverages the ability to replace a piece, which it does to the moving piece by replacing it with an opponent pawn. As the opponent's pawn can't put its king in check, if the board remains in check after this replacement then the moving piece was not the cause of the initial check - it was a Discovered Check.

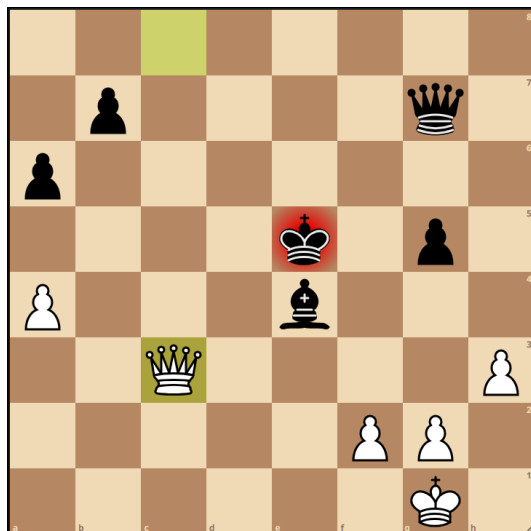
4.2.4 Skewer

A Skewer is an attack upon two pieces in a line and is similar to a Pin. A Skewer is sometimes described as a "Reverse Pin"; the difference is that in a Skewer, the more valuable piece is in front of the piece of lesser value [56]. If the piece being attacked is not a king, then it is a Relative Skewer, otherwise, it is an Absolute Skewer. Examples of both can be seen in Fig. 4.7.

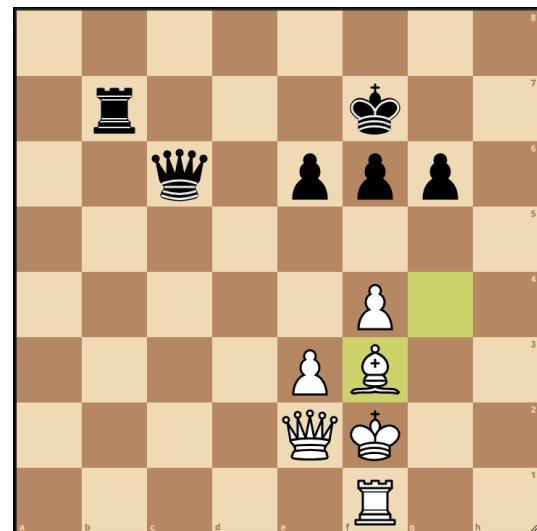
A high-level description of the TRA is given in Alg. 7. For a move to be considered a Skewer, it must fulfil the following criteria:

- The square moved to is undefended
- The first piece of the Skewer is higher-valued than the piece Skewering
- It is a new Skewer

Like Pins, a Skewer can be revealed rather than made directly by the moving piece. The algorithm is straightforward, reusing the helper functions Alg. 10 and Alg. 11 to find and check for existing x-rays. Additionally, a function for checking if the king can defend the x-rayed piece in an Absolute Skewer is given in Alg. 8. The lengthy conditional ensures the relative-value logic of two kinds of Skewer.



(a) Absolute Skewer



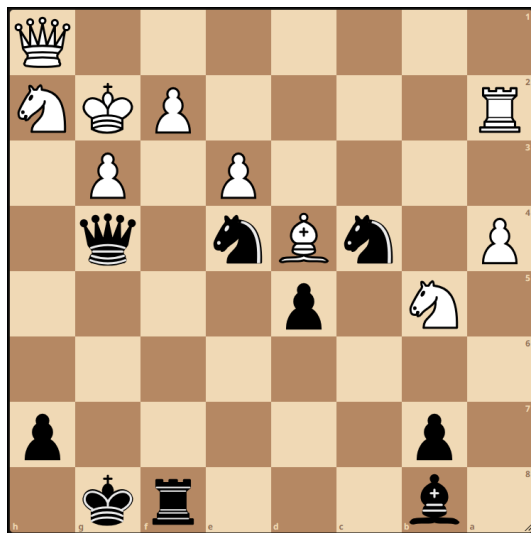
(b) Relative Skewer

Figure 4.7: Examples of Skewers

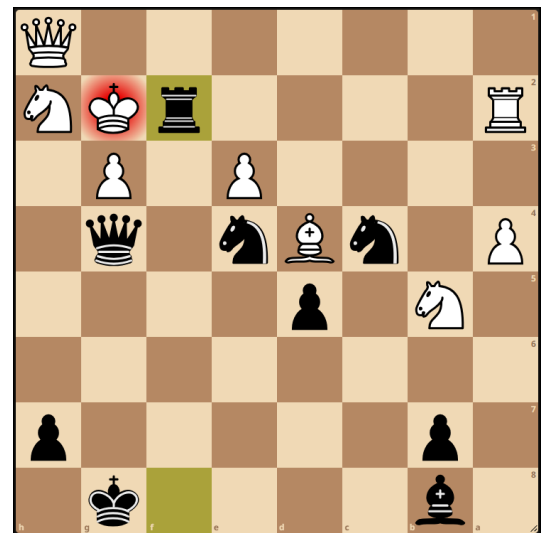
4.2.5 Undermine (Remove the Defender)

An Undermine (or removing the defender) happens when a player captures a piece that has been performing a defensive role. The removed piece could be defending other pieces or squares. After eliminating that piece, the attacking player can take advantage of the positional weakness created by the absence of that defender [57]. An example is given in Fig. 4.8, whereby the pawn on f2 is preventing the black queen from giving a devastating check on g3 - the best sequence of moves is to capture the pawn on f2 with a check, forcing white to respond, which forces a "Mate in 4" after *Rxf2, Qxg3, Kf1, Qxf2#*.

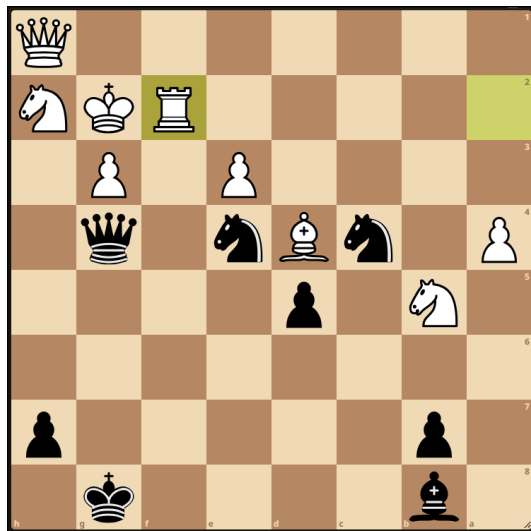
A high-level description of the TRA is given in Alg. 4.8. This is the first TRA which requires a sequence of moves rather than just an individual move. This implementation extracts instances of the tactic whereby each move in the sequence was a capture. I consider this the absolute version of the Undermine, where there is a clear exchange of pieces with the final capture being a consequence of the initial capture.



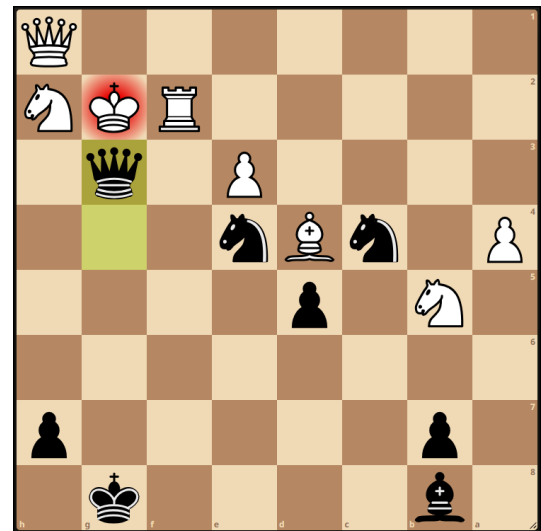
(a) Pawn on f3 is Defending g2



(b) Remove the Defender



(c) Forced Recapture



(d) Devastating Check

Figure 4.8: Example of an Undermine

4.3 Data Preparation

The next phase of this project involved choosing a selection of games from the LiChess database for analysis. This process aimed to ensure a consistent and unbiased selection of games across different time controls and player skill levels. To achieve this, I ensured that the chosen games were valid and that active players played them.

4.3.1 Which Games are Valid?

There are millions of games in the LiChess database spanning several years. From ranked games between grandmasters to friendly non-competitive games, there is no shortage of variety in analysing whichever kind of game you want. For this analysis, I chose games from August 2023 as I deemed the 95 million games played sufficient. The following criteria must be met for a game to be valid:

- The number of moves played was greater than eight
- It was a ranked game
- It was one of the following game variants: Rapid, Blitz, Bullet
- It was a regular time control
- The game's PGN was not compromised

Several criteria were established to determine the validity of games from the LiChess database to ensure the integrity of the dataset. Firstly, games with a minimum of eight moves were considered to ensure they progressed beyond the initial stages and potentially entered tactical territory. One move consists of a turn by each player, and each turn (ply) is called a half-move. Additionally, only ranked games were included to maintain a competitive environment for analysis. Games were restricted to the Rapid, Blitz, and Bullet variants, with Rapid games typically lasting between 20 to 60 minutes, Blitz games between 6 to 25 minutes, and Bullet games lasting 1 to 5 minutes. This selection ensured coverage of a range of time controls commonly used in competitive play, each presenting distinct strategic challenges and opportunities for tactical manoeuvring. Furthermore, only games played under regular time controls were chosen to maintain consistency across the dataset and avoid analysing games with irregular time controls that might introduce confounding factors. Finally, games with compromised Portable Game Notation (PGN) files, often due to invalid timestamps or other formatting issues, were excluded to ensure data reliability and consistency throughout the analysis process.

4.3.2 How is a Player Deemed Active?

Determining the activity level of players in the dataset is crucial for understanding player engagement and participation. The criteria for deeming a player active varied based on the game type, reflecting the differing playing habits associated with games of varying lengths. As illustrated in Figure 4.9, players tend to engage in more games when the game duration is shorter.

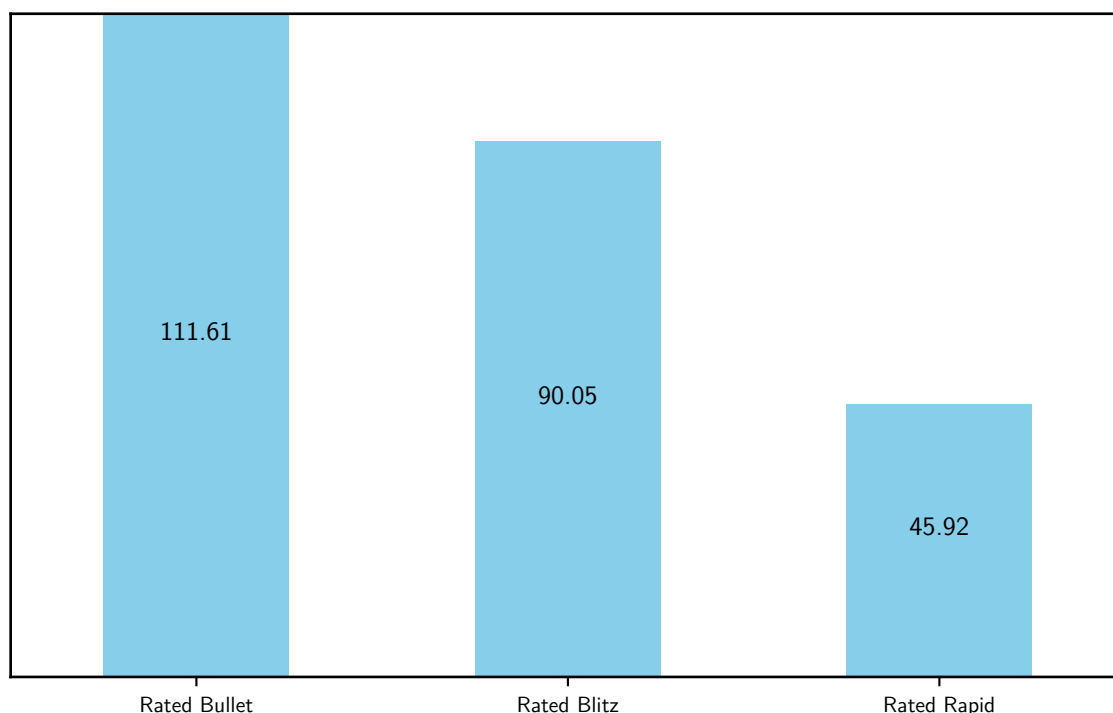


Figure 4.9: The average number of games played by players during August 2023

To identify active players, the following steps were undertaken. Firstly, games were filtered based on the first three criteria of game validity, as the remaining two criteria did not directly impact a player's activity level. Subsequently, the number of games played by each player was counted, and players who played less than five or more than 1000 games were excluded. This range was selected to filter out players who either played infrequently or engaged in an unfeasibly large number of games, potentially indicating shared accounts or usage for testing chess engines, exemplified by accounts named after chess engines such as Maia, Leela, and Stockfish present in the dataset.

Following this filtering process, a player was deemed active if they scored above the 50th percentile of games played for the respective game variant. This approach resulted in the identification of 786,894 active players in August 2023. Table 4.1 provides summary statistics of games played by players, categorised by game variant. These statistics offer insights into the distribution of player engagement across different time controls, revealing that active players average at least a game daily in the shorter variants and at least two games every three days in Rapid.

Variant	Count	Mean	Std	Min	25%	50%	75%	99%
Rated Blitz	859447	90.05	117.83	5	13	40	119	673
Rated Bullet	441653	111.61	158.38	5	13	40	141	836
Rated Rapid	500385	45.92	61.11	5	9	20	55	388

Table 4.1: Summary Statistics of Games Played by Players

4.3.3 Game Selection

Following on from the work outlined in the previous subsections, here are the remaining steps of the data preparation process:

- **New Account Removal:** Exclude games from new accounts.
- **Elo Rating Assignment:** Assign a rating to each game.
- **Elo Grouping and Sampling:** Group games by Elo and select representative samples.
- **Game Cleaning:** Remove compromised PGN files and games with irregularities.

The initial step in the game selection process involved filtering for games played by the identified active users. This ensured that the dataset comprised games played by participants who were likely to exhibit strategic gameplay and encounter tactical motifs consistently. Next, the identification and removal of games played by new accounts were carried out to mitigate potential biases introduced by inexperienced or unreliable players. Once the active users' games were identified, Elo ratings were assigned to each game based on the average of the two players' ratings, providing a standardised measure of player skill for analysis. This allowed for meaningful comparisons and insights into the relationship between player skill and tactical play.

Once games were grouped by Elo classification for each game variant, representative samples were selected to ensure statistical significance while maintaining dataset diversity. Random subsamples comprising 20% (10,000 games) of each Elo group were chosen to prevent overpopulation by the most active users. This approach aimed to maintain a balanced representation of player engagement across different skill levels, facilitating comprehensive analysis of tactical motifs in chess without bias towards highly active users. Following the selection of samples, a cleaning process was undertaken to remove compromised PGN files; games with irregular time controls or termination types. This ensured data reliability and consistency, minimising potential sources of bias in the dataset.

Completion of the cleaning process meant that the dataset was ready for analysis. It comprised a representative sample of games played by active users across different Elo classifications and game variants, providing a robust foundation for exploring tactical play in chess.

4.4 Tactic Extraction

This crucial phase of the project involved pairing the Tactical Recognition Algorithms (TRAs) with Stockfish to analyse hundreds of thousands of chess games for tactical motifs. Stockfish evaluates positions using advanced algorithms and heuristics to determine the best moves. It considers factors beyond just material advantage, such as piece activity, pawn structure, king safety, and positional advantages. The evaluation is represented numerically in *Centipawns*, where one Centipawn is equivalent to one-hundredth of a pawn and a higher score indicates a more advantageous position. The evaluation also depends on the depth, time, and number of threads available for the engine. A higher depth, more time and more threads will provide a better evaluation, which is crucial for identifying the best moves and determining their impact on the game's outcome.

Stockfish evaluated certain positions of each game, providing the best sequence of moves, referred to as the *"top engine line"*. The TRAs searched for tactical motifs in the top engine line and if this line was not played in the game, an additional search was conducted on the player's line if it was still competitive. The player's line was considered competitive if its evaluation was within 10% of the top engine line's evaluation. Alg. 12 provides a high-level description of this process, where the input consists of the selection of games chosen previously. The algorithm iterates through each game, analysing positions beyond the 12th half-move until either the game terminates or exceeds the 79th half-move. This lower bound avoids analysing opening theory, while the upper bound is based on the average number of half-moves a chess game lasts for [58]. Setting this upper bound is essential to ensure consistent analysis and to filter out moves played when the game is essentially over, but players continue non-competitively.

Column Name	Type	Description
game_id	Int	The unique identifier of the game
found	Int	1 if the tactic was played in the game, else 0
top_engine_move	Int	1 if it was the top engine move, else 0
type	String	The name of the tactic
piece	String	The piece moved
fen	String	The FEN representation of the previous position
move	String	The move to be played
sequence	List	A list of moves to be played (for Undermines)
player_elo	Int	Player's elo
game_elo	Int	The elo of the game
time_control	String	The time control of the game
ply	Int	Half-move number
move_time	Int	Duration of move (seconds)
clock_time	Int	The time remaining in the position
is_winning	Int	1 if the tactic is game-defining else 0

Table 4.2: Tactics DataFrame Structure

Once a tactical motif was identified, it needed to be stored in a structured manner. A Pandas DataFrame was employed for this purpose, seamlessly integrating with the environment setup. Each entry in the DataFrame represents a tactical instance, with the column labels and descriptions

provided in Table 4.2. Notably, the final column specifies if the tactical instance was *"game-defining"*, indicating a tactic that significantly alters the game's outcome.

Determining whether a tactical motif is "game-defining" is crucial for understanding its impact on the game's outcome. Alg. 13 outlines a systematic approach to classify moves based on their significance. By comparing the evaluation scores of the two top engine lines, the algorithm identifies if the second-best move results in a substantial advantage for the opponent. This indicates a *"blunder"* by the player, as the best move significantly alters the game's outcome. The threshold for defining a move as "game-defining" is set at a Centipawn loss greater than 300, equivalent to an advantage of the relative value of a bishop or knight. This approach aligns with a common understanding in chess, where blunders are characterised by a significant positional disadvantage resulting from a single half-move. By employing this algorithm, the analysis focuses on tactical motifs that are not only present but also game-changing.

Chapter 5: Results

In this chapter, we delve into the findings from the analysis of tactical motifs in chess games. This chapter is structured to address the key research questions posed previously, offering insights into the role of tactical proficiency, its correlation with skill level, the efficacy of different tactics, and the impact of time on tactical performance. We begin by examining the impact of tactical motifs on gameplay across different skill levels and game variants. Subsequently, we explore the correlation between tactical proficiency and skill level, analysing trends in the identification and execution of game-defining tactics. Following this, we investigate which tactical motifs would be most beneficial for players to focus on, providing insights into the frequency and effectiveness of various tactics across different Elo levels. Finally, we analyse the effect of time on tactical proficiency, examining how players' performance varies with the time remaining in the game. Through this structured approach, we aim to provide a comprehensive understanding of the role of tactics in chess gameplay.

5.1 How often do tactics appear in gameplay?

The first question posed in our investigation pertains to the significance of tactical motifs in chess. To address this query, we delve into an analysis of tactical occurrences across different Elo levels and game variants. Our examination is supported by visualisations that offer insights into the frequency and impact of tactical motifs throughout various levels of play.

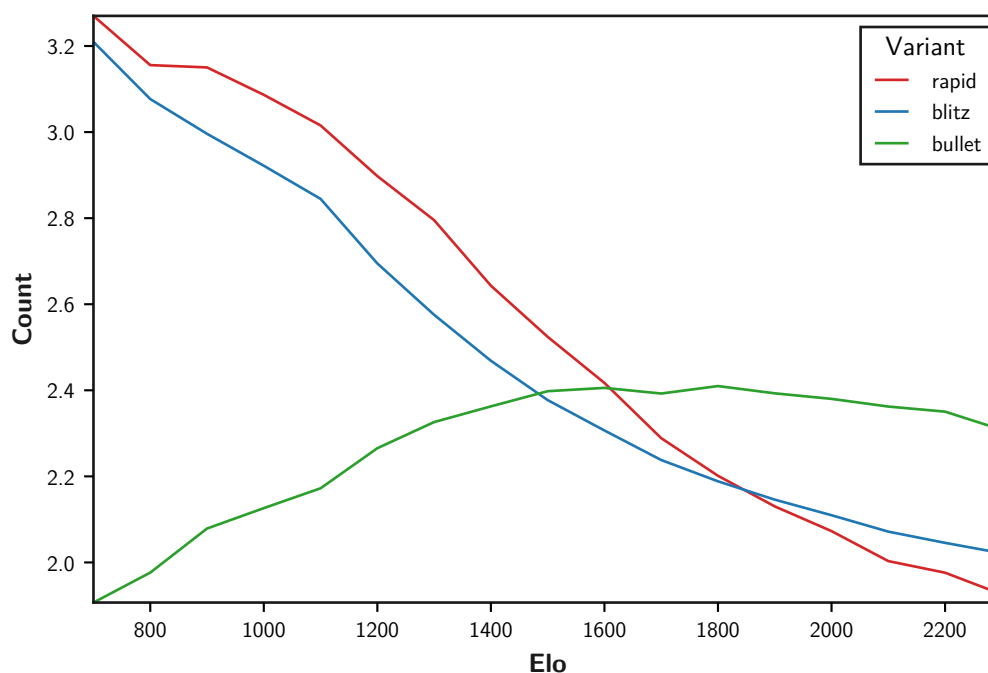


Figure 5.1: Count per game of tactics across Elo and by Variant

The initial visualisation, depicted in Fig. 5.1, portrays the average number of tactics per game plotted against players' Elo ratings, stratified by game variants: Rapid, Blitz, and Bullet. Notably,

we observe divergent trends among the three variants. In Rapid and Blitz games, the number of tactics per game steadily declines as Elo ratings increase, starting at approximately 3.2 tactics per game for players with an Elo of 700 and gradually decreasing to just above 2 tactics per game for players with an Elo of 2300. However, in Bullet games, the trend differs significantly. Here, we observe a steady incline in the number of tactics per game until around 1800 Elo, after which it begins to decline gradually. This disparity in trends can be attributed to the considerably lower average number of moves in Bullet games compared to Rapid and Blitz, as seen in Fig. 5.2, resulting in a comparatively lower number of tactics per game at lower Elo levels in the Bullet variant. Notably, the average number of moves per game across the variants converge at advanced skill levels, at an Elo of approximately 1900.

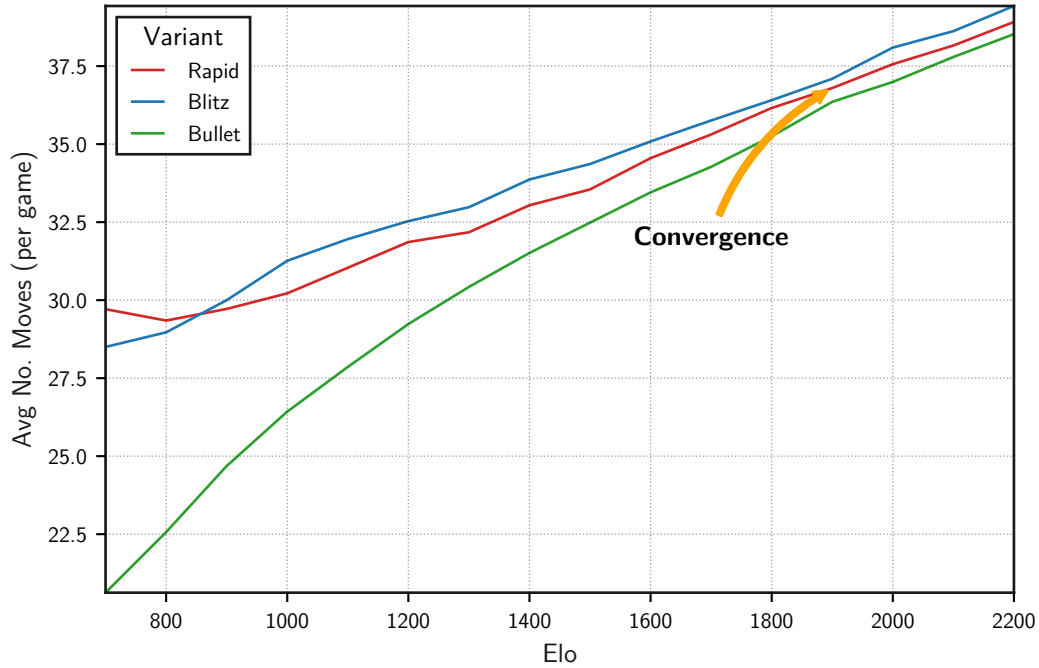


Figure 5.2: Average number of moves per game across Elo by Variant

Further exploration into the impact of tactical motifs is facilitated by examining the occurrence of *game-defining* tactics. The corresponding visualisation, Fig. 5.3, mirrors the previous trend observed for total tactics, albeit with a notable distinction. Despite a decrease in overall frequency, game-defining tactics remain *consistently proportional* to the total number of tactics, comprising around 11-13% of total tactics across all elo levels and game variants. This finding suggests that while tactical opportunities may diminish at higher skill levels, the potential for game-altering moves remains palpable, offering players the chance to seize significant advantages during gameplay.

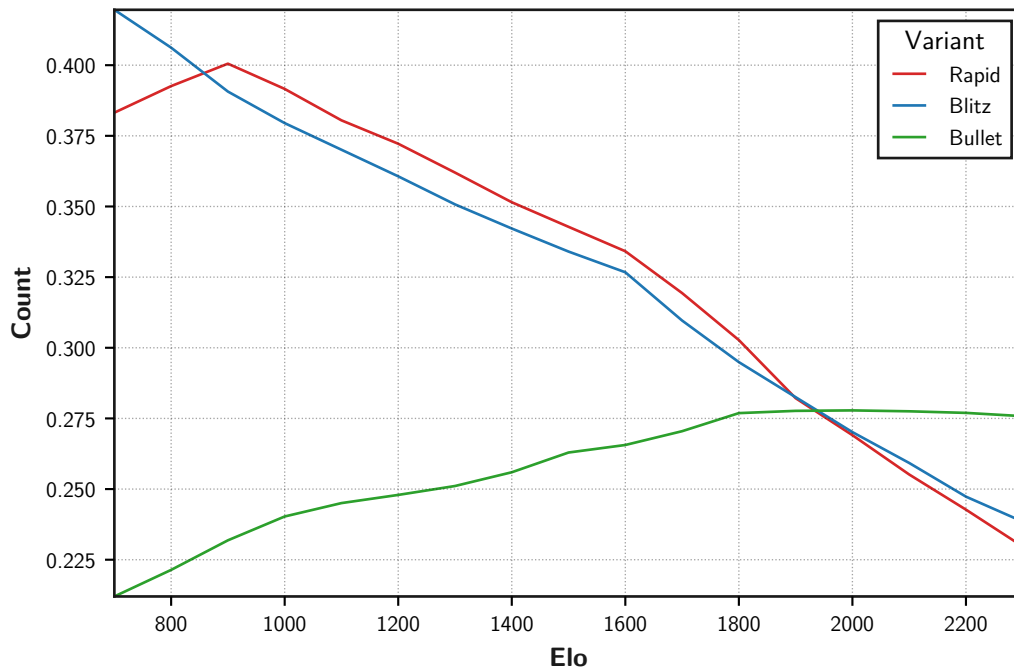


Figure 5.3: Count per game of game-defining tactics across Elo by Variant

A more granular analysis of tactical occurrences is provided through Fig. 5.4, detailing the frequency of each unique tactic across game variants. Notably, Forks emerge as the most prevalent tactic across all variants, followed closely by Pins. Interestingly, the total counts of each tactic exhibit similarity across variants. Furthermore, the breakdown of Forks into subtypes- Forks dealt by knights, queens, or the other pieces-adds depth to our analysis, highlighting the nuanced nature of tactical engagements in chess.

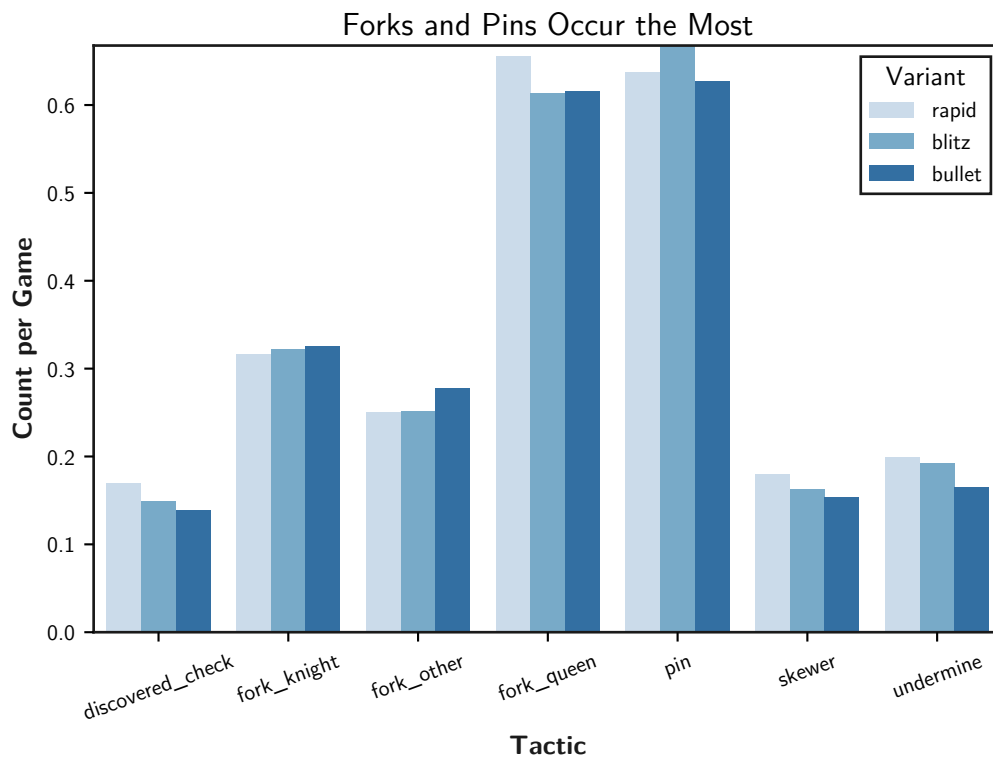


Figure 5.4: Count per game of each tactic by Variant

In conclusion, our analysis demonstrates that tactical motifs impact gameplay across all skill levels and game variants. While the frequency of tactical occurrences may diminish with increasing player skill, the potential for game-defining moves remains ever-present, offering players the opportunity to secure both marginal and decisive advantages over their opponents. These findings underscore the strategic importance of tactical proficiency in the realm of chess gameplay.

5.2 Does tactical proficiency correlate with skill?

The next question under examination delves into the correlation between tactical proficiency and skill level in chess. To address this inquiry, we look at two visualisations aimed at measuring the ability of players to identify and execute tactical motifs across different Elo levels and game variants.

The first visualisation, depicted in Fig. 5.5, illustrates tactical proficiency across Elo levels for each game variant: Rapid, Blitz, and Bullet. Tactical proficiency is measured as the percentage of correctly identified and executed tactics out of the total opportunities presented. Notably, all variants exhibit an increasing trend in tactical proficiency as Elo ratings rise, with Rapid demonstrating the most significant improvement. Starting at 52% proficiency for players with an Elo of 700, the proficiency level climbs steadily to just under 68% for players with an Elo of 2300. This trend underscores a clear correlation between tactical proficiency and skill level, as higher-rated players demonstrate an enhanced ability to identify and execute tactical motifs.

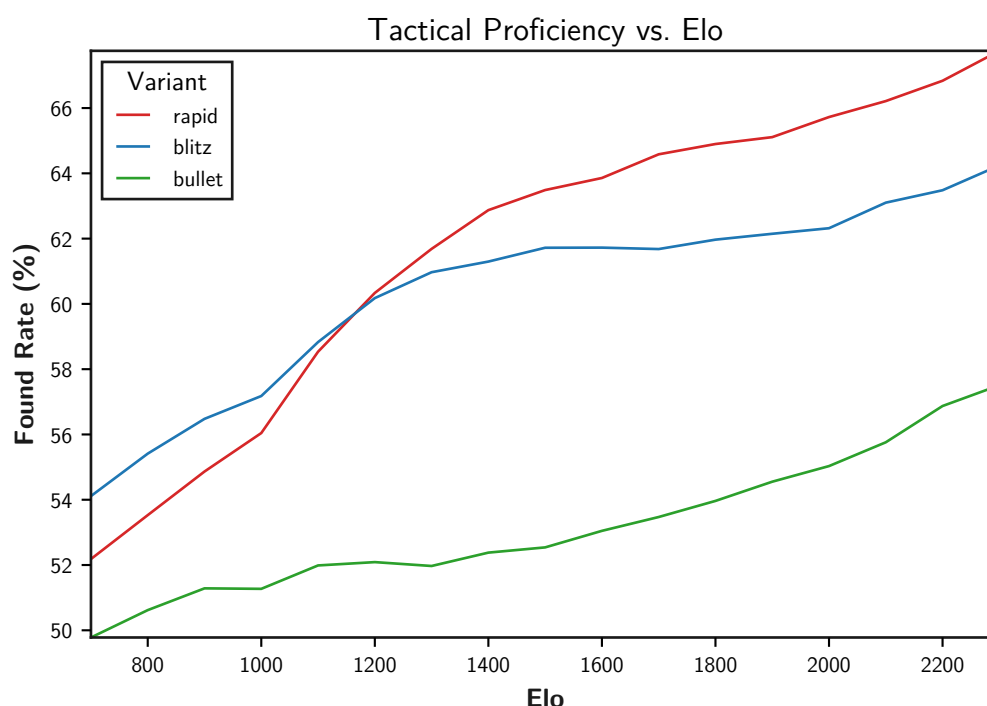


Figure 5.5: Found rate of tactics across Elo by Variant

Fig. 5.6 explores the tactical proficiency of game-defining versus non-game-defining tactics across Elo levels and game variants. Here, tactical proficiency is again measured as the percentage of correctly identified and executed tactics out of the total opportunities presented, with a focus on game-defining occurrences. Across all Elo levels and game variants, the tactical proficiency of game-defining occurrences surpasses that of non-game-defining tactics. Notably, there is a greater rate of improvement in proficiency for game-defining occurrences, further reinforcing the

correlation between tactical proficiency and skill level. Rapid again stands out, exhibiting the greatest improvement in proficiency, soaring from 56% to just under 80%.

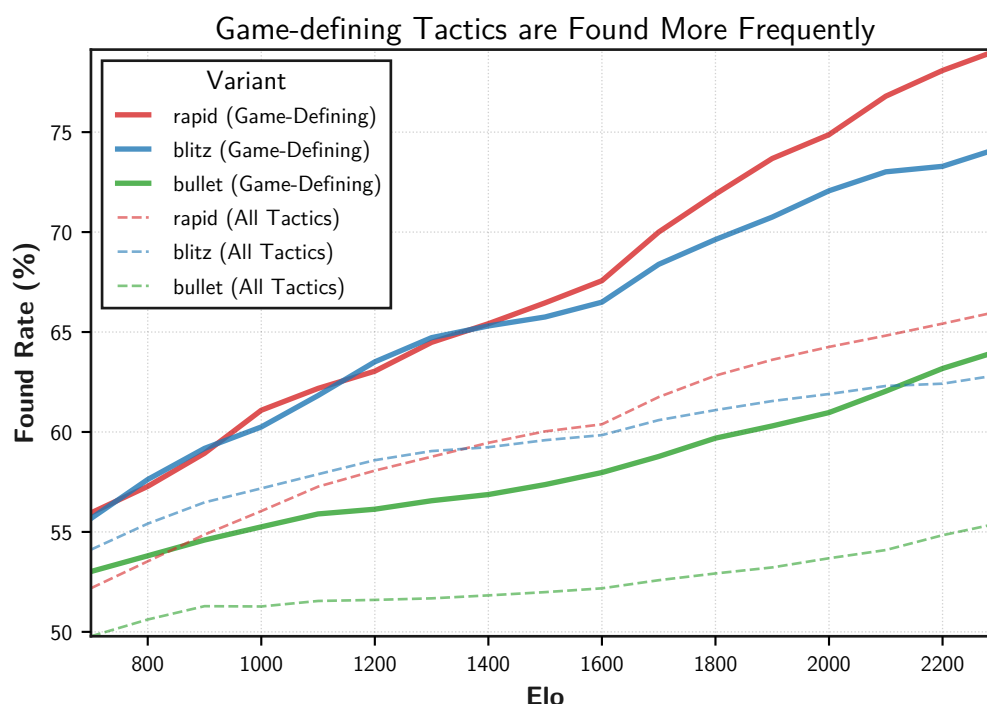


Figure 5.6: Found rate of game-defining tactics across Elo by Variant

In conclusion, our analysis reveals a clear correlation between tactical proficiency and skill level in chess. As players ascend the Elo ladder, their ability to identify and execute tactical motifs improves significantly, with game-defining occurrences demonstrating a particularly strong association with skill level. These findings underscore the strategic importance of tactical proficiency in determining success in chess gameplay.

5.3 Which tactics should players focus on?

The subsequent inquiry in our investigation centres on identifying which tactical motifs players could benefit most from improving. To address this question, we analyse a series of visualisations aimed at measuring the proficiency and impact of various tactics across different Elo levels.

The first visualisation, depicted in Fig. 5.7, illustrates the tactical proficiency across Elo levels for each tactic in Blitz games. Here, we measure the ability of players to identify and execute each tactic, across the Elo range of 700 to 2500. Notably, all tactics exhibit an increasing trend in proficiency as Elo ratings rise. However, the one multi-move tactic, the Undermine, stands out with significantly lower proficiency levels across all Elo levels but demonstrates the largest improvement over the Elo range. This suggests that Undermines presents a greater challenge for players to identify and execute effectively compared to other tactics.

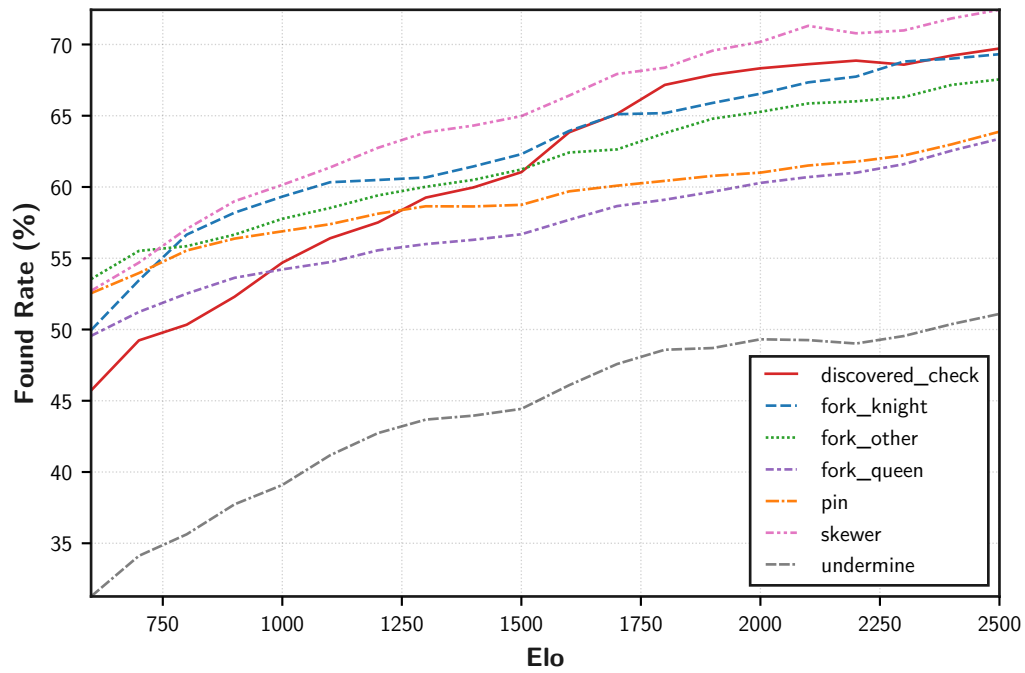


Figure 5.7: Found rate of each tactic across Elo in Blitz

Fig. 5.8 explores the tactical proficiency across Elo levels for each type of Fork, distinguished by the forking piece. By measuring the ability of players to identify and execute different Forks, we find that Knight Forks exhibit the *highest proficiency* and show the greatest improvement across Elo levels. This suggests that Knight Forks are more readily identified and executed by higher-rated players, highlighting their strategic significance in gameplay.

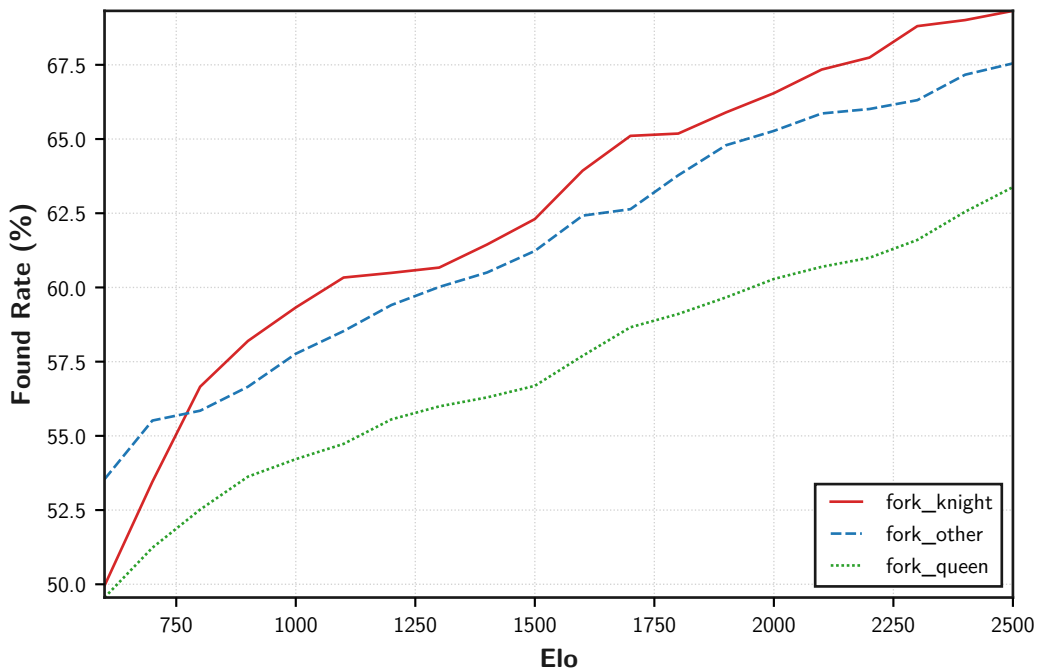


Figure 5.8: Found rate of each Fork subtype across Elo in Blitz

Fig. 5.9 presents the count per game of each tactic across Elo levels in all the games. While most tactics exhibit a decreasing trend in occurrence as Elo ratings increase, Undermines and Pins demonstrate an *increasing trend*. This implies that there are more opportunities to identify and

execute Undermines and Pins at higher Elo levels, indicating their strategic importance in gameplay. Lastly, the final visualisation, depicted in Fig. 5.10 illustrates the proportion of game-defining tactics broken down by tactic. Notably, Skewers emerge as the most *devastating tactic*, with the largest proportion of game-defining occurrences, followed by Knight Forks. This underscores the potent threat posed by Skewers in gaining a sizable advantage over opponents. Pins and Undermines have the smallest proportion of tactics that are game-defining, highlighting that they are more frequently employed to gain marginal advantages over opponents.

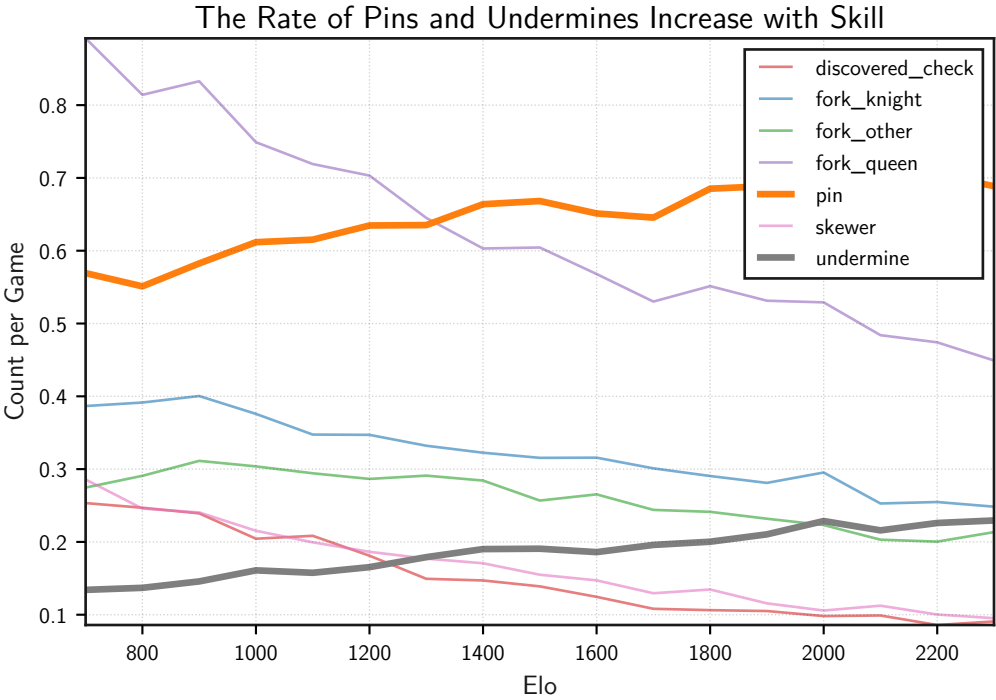


Figure 5.9: Count per game of each tactic across Elo by Variant

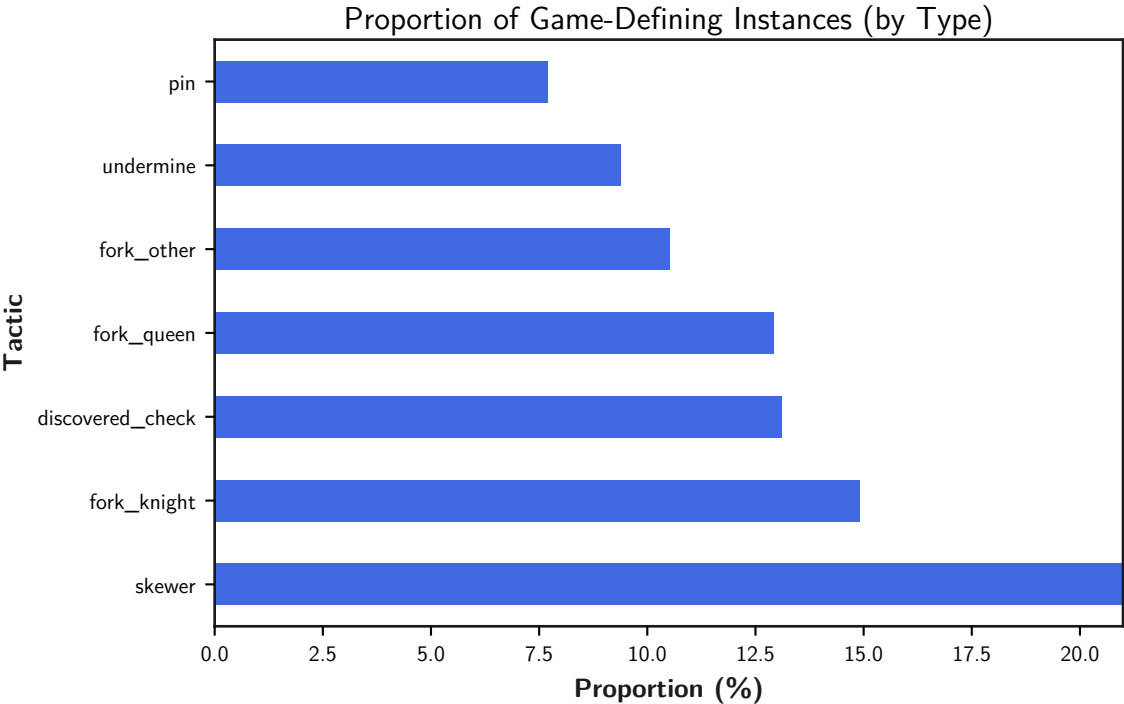


Figure 5.10: Proportion of game-defining instances by tactic

In conclusion, our analysis suggests that improving proficiency in Forks, particularly Knight Forks, would be *most beneficial* to players across all tactics. Despite the decreasing instance rate at higher Elo levels, Forks remain the most present tactic and provide a potent threat, especially in gaining significant advantages. Additionally, Pins exhibit increasing presence at higher Elo levels and are utilised more to gain marginal advantages. Skewers, on the other hand, represent a devastating tactic, particularly at lower Elo levels where its presence is more felt. Meanwhile, the Undermining tactic, although least proficient, shows the largest improvement and increased presence, indicating active efforts by higher-level players to identify and execute this tactic effectively.

5.4 How Does Time Affect Tactical Proficiency?

The final question in our exploration delves into the impact of time on tactical proficiency in chess gameplay. To address this inquiry, we analyse a series of visualisations aimed at examining the relationship between time and tactical performance across different game variants and Elo levels.

The first visualisation, depicted in Fig. 5.11, illustrates the average move time of found tactics across Elo levels for each game variant: Rapid, Blitz, and Bullet. Notably, there is a *sizable gap* in average move times across variants, with Rapid games exhibiting the longest average move times, followed by Blitz and then Bullet. Additionally, *diverging trends* emerge: while Rapid's average move time increases at higher Elo levels, the move times in Blitz and Bullet games decrease. This divergence suggests that the time control of the game significantly influences the average move time of tactical manoeuvres. Specifically, Rapid players may have the luxury to calculate more moves ahead due to the increased time available, and as players improve, they are capable of calculating longer sequences, resulting in longer move times.

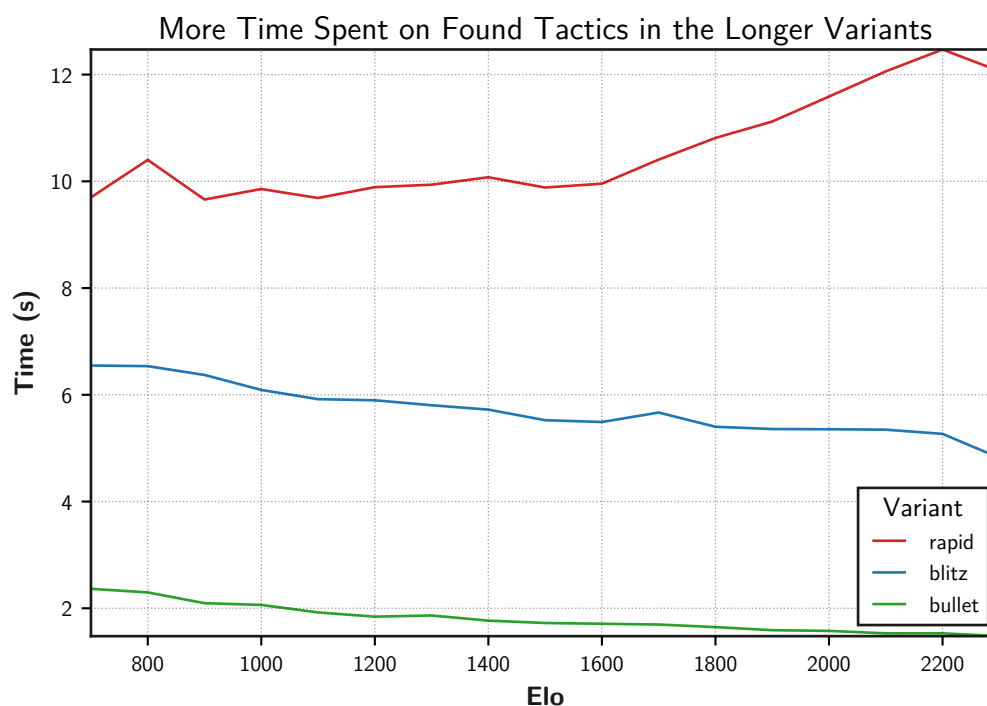


Figure 5.11: Average move time of played tactics across Elo by Variant

Fig. 5.12 explores the tactical proficiency versus the time remaining for each game variant. Across all variants, there is a consistent trend of increased proficiency with less time remaining until

approximately 45% of the starting time remains, at which point proficiency begins a rapid decline. Notably, Blitz players outperform Rapid, and Rapid players outperform Bullet players. This finding suggests that Blitz players have a *higher tactical proficiency*, potentially indicating superior skill levels. Interestingly, as time dwindles, players struggle to maintain proficiency, particularly in Blitz and Bullet games where low proportions of time remaining equate to mere seconds.

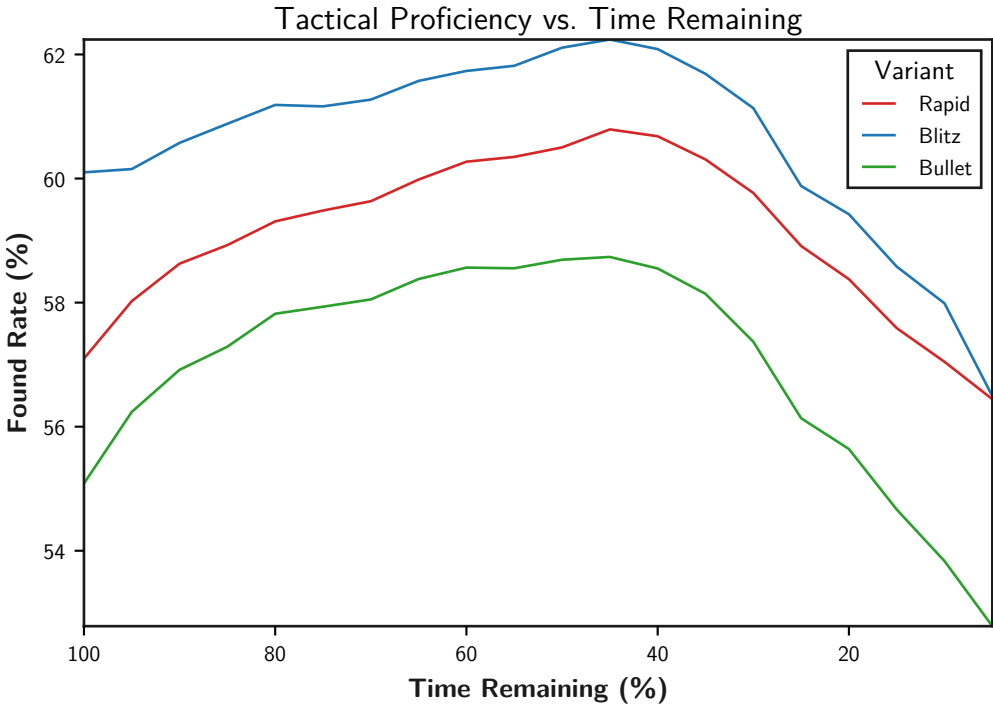


Figure 5.12: Found rate across the proportion of time remaining by Variant

The remaining visualisations, Fig. 5.13, Fig. 5.14, Fig. 5.15, present the distribution of move times in positions with a tactical motif present and tactical proficiency versus move time intervals for each game variant. Each dual figure consists of a histogram depicting the distribution of move times and a bar chart illustrating tactical proficiency versus move time intervals.

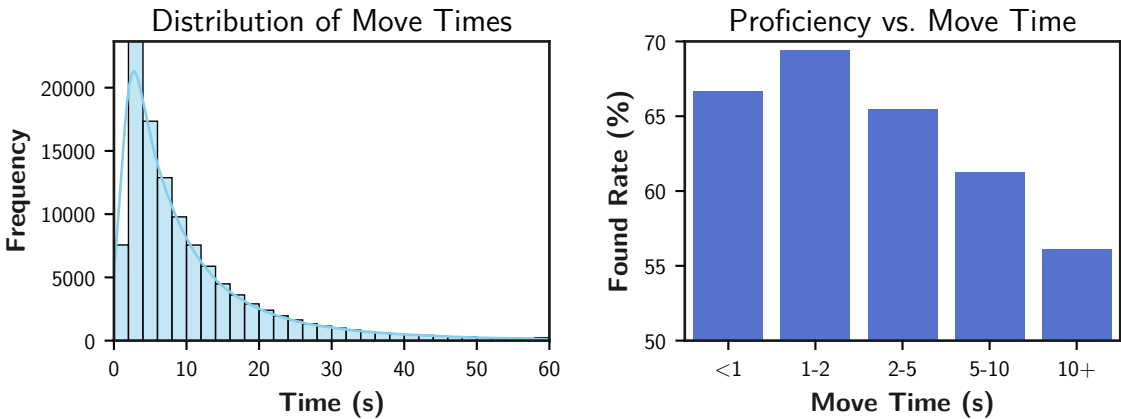


Figure 5.13: Move Time Insights of Tactical Positions in Rapid

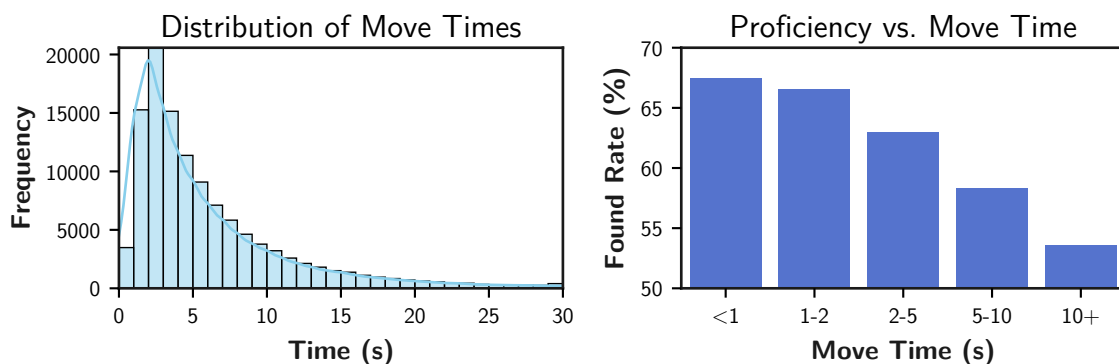


Figure 5.14: Move Time Insights of Tactical Positions in Blitz

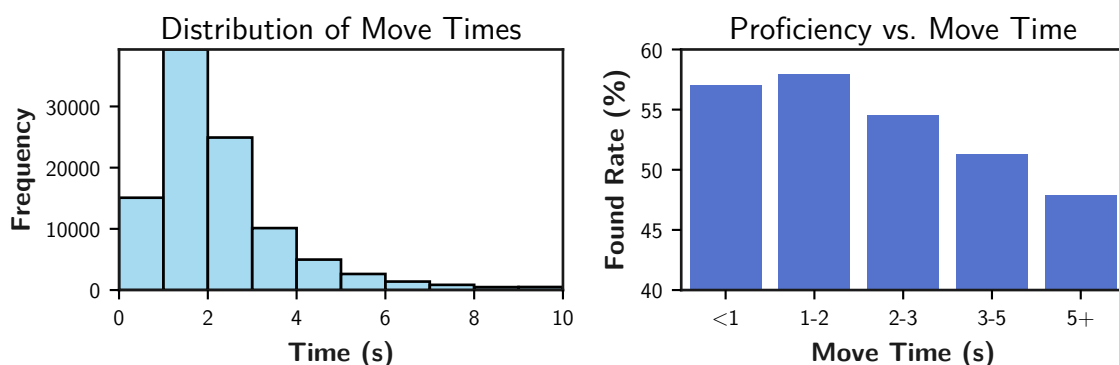


Figure 5.15: Move Time Insights of Tactical Positions in Bullet

Across all variants, the move time distributions exhibit similar right-skewed patterns, indicating commonalities in time management strategies among players. Surprisingly, moves made in less than 2 seconds demonstrate the *highest proficiency* across all variants, while proficiency sharply declines for moves that take 5 or more seconds. This unexpected finding suggests that proficiency is associated more with pattern recognition and move planning than calculation duration.

In conclusion, the analysis reveals that the amount of time remaining in games has a profound effect on tactical proficiency. Players struggle to maintain proficiency as their game clock deteriorates, with the highest proficiency observed when moves are made in a matter of seconds. These findings provide valuable insights into the dynamics of time management and tactical performance in chess.

5.5 Conclusion

In conclusion, our exploration into the realm of tactical motifs in chess has unearthed several key insights. Firstly, we found that tactical proficiency impacts gameplay across all skill levels and game variants, with ample opportunities to gain advantages from tactical manoeuvres. Secondly, we observed a clear correlation between tactical proficiency and skill level, with players demonstrating greater proficiency as their Elo ratings increased. Thirdly, our analysis highlighted the varied benefits of improving proficiency in specific tactics, with Forks emerging as particularly advantageous due to their prevalence and potency. Lastly, we uncovered the profound influence of time on tactical proficiency, revealing that players struggle to maintain proficiency as their game clock diminishes. These findings underscore the intricate dynamics of chess gameplay and provide valuable insights for players, coaches, and enthusiasts alike.

Chapter 6: Conclusion

In this chapter, we provide a comprehensive overview of the key findings and milestones achieved throughout the project. We begin with a summary of the steps taken and milestones reached, followed by an examination of the challenges faced and limitations encountered. Lastly, we conclude with remarks on the success of the project and its implications for future research and development in the field of chess expertise.

6.1 Summary

Throughout this project, I undertook a comprehensive journey into the realm of chess analysis, aiming to uncover valuable insights into tactical motifs and their significance in the game. This endeavour began with a literature review, allowing me to gain an understanding of the theoretical foundations and previous research findings in this domain. Armed with this knowledge, I proceeded to set up the project environment, familiarising myself with the intricacies of the Python-Chess package and Stockfish engine, essential tools for conducting the analysis.

The exploration of the LiChess database provided a wealth of data, comprising millions of chess games played over several years. Through careful consideration and refinement, I established criteria for selecting games for analysis, ensuring a consistent and unbiased sample across different time controls and player skill levels. With the criteria in place, I selected games that met the specified requirements, laying the groundwork for analysis.

Cleaning and partitioning the selected games for analysis marked a significant milestone in the project. This process involved preprocessing the data to remove ineligible games and extracting individual tactical positions using the implemented Tactical Recognition Algorithms (TRAs). Successfully deploying the TRAs enabled the extraction of over 300,000 tactical positions from the selected games, providing a rich dataset for analysis.

The analysis phase yielded novel insights into various aspects of tactical motifs in chess. Through rigorous examination, I uncovered patterns in the frequency of tactics, revealing intriguing trends across different skill levels and game variants. Furthermore, I investigated the correlation between tactical proficiency and skill level, shedding light on the importance of tactical prowess in chess mastery. Additionally, I identified specific tactics that players should focus on to enhance their gameplay, offering practical guidance for skill improvement. Finally, I delved into the effect of time on tactical proficiency, unveiling the intricate relationship between time management and strategic decision-making in chess.

In summary, this project has been a journey of discovery, culminating in valuable insights into the world of chess analysis. The successful implementation of TRAs, coupled with the thorough analysis of extracted tactical positions, has advanced our understanding of tactical motifs and their role in the game. Moving forward, these findings pave the way for future research and development in chess analysis and training, offering new opportunities for innovation and growth within the chess community.

6.2 Challenges and Limitations

Challenges: Familiarising myself with the Python-Chess package proved to be a significant challenge, despite my proficiency in the Python programming language. The extensive documentation of the package required a considerable amount of time and effort to navigate effectively, delaying the initial stages of the project.

Integrating Stockfish with the Python-Chess package presented another hurdle, as it was my first experience with a chess engine. Understanding the intricacies of how these two components interacted and functioned together required thorough research and experimentation.

Designing and implementing the Tactical Recognition Algorithms (TRAs) posed a unique challenge, as there were limited resources available for reference. With no established guidelines or prior examples to follow, I relied heavily on my intuition and logic to develop these algorithms. While the Python-Chess package provided some helpful functionality, the majority of the implementation was based on independent problem-solving.

The computational cost of tactic extraction was a persistent challenge throughout the project. Balancing the trade-off between time, depth, and the number of nodes used by Stockfish was essential to optimise the efficiency of tactic extraction. Striking a harmonious balance allowed for a sufficient number of games to be analysed without compromising the quality of evaluations.

Limitations: One notable limitation of our approach is the exclusion of master-level players from the analysis. While this decision was made to focus on a broader range of skill levels, it may have overlooked potential insights that could be gleaned from studying advanced tactics employed by top-tier players.

Our methodology only accounted for the identification of a single tactic per position during the extraction phase. This oversimplification fails to capture the complexity inherent in chess positions, where multiple tactical motifs may coexist simultaneously. Future iterations should consider identifying and extracting multiple tactics within a single position to provide a more comprehensive analysis.

Our analysis primarily focused on a handful of basic tactical motifs, neglecting more advanced tactics prevalent in higher-rated games. This narrow scope may explain the observed decrease in the frequency of tactics in higher-rated games and underscores the need for a more comprehensive taxonomy of tactical motifs to encompass a broader range of gameplay scenarios.

6.3 Future Work

Despite these limitations, our work lays a solid foundation for future research and development in the field of chess analysis and training. One promising avenue for future exploration is the development of an AI-powered training assistant tailored specifically to enhance tactical proficiency in chess players.

By leveraging the exclusive data on tactical proficiency generated through our analysis and access to thousands of tactical positions, this assistant could offer personalised training recommendations to users. Through continuous analysis of tactical gameplay in users' games, proficiency could be tracked and displayed, allowing for targeted improvement strategies. Moreover, this assistant could expand its offerings beyond tactical training to include chess puzzles, positional theory, and more,

all informed by the rich dataset accumulated during our analysis. The potential to monetise such a service could open up new opportunities for innovation and growth within the chess community.

In summary, while our current analysis represents a significant step forward in understanding tactical motifs in chess, there remains ample room for future exploration and innovation. By addressing the identified limitations and embracing new opportunities, we can continue to push the boundaries of chess analysis and training, ultimately empowering players to reach new heights of strategic mastery.

6.4 Concluding Remarks

In conclusion, this project has been a resounding success, culminating in novel insights into the role of tactics in chess skill. The exploration of tactical motifs and their correlation with player proficiency has opened doors to a deeper understanding of chess gameplay dynamics.

The realm of chess research is undoubtedly exciting, with endless possibilities for further exploration and discovery. As the field continues to evolve, I am eager to witness further developments in understanding the intricate interplay between tactics and overall chess skill.

I am hopeful that my work can serve as a foundation for future research endeavours. By shedding light on the nuances of tactical proficiency, I aspire to inspire further investigations that delve deeper into the complexities of chess gameplay. I look forward to seeing how my findings contribute to the broader body of knowledge in this domain and how they can be leveraged to enhance our understanding of chess strategy and skill development.

Acknowledgements

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Algorithms and Code

Code Repository: <https://csgitlab.ucd.ie/byron.ogorman/chess-tactics-analysis>

Algorithm 1 Fork TRA

Require: *board* and *move*

Ensure: True if the move is a Fork, False otherwise

attacking_square $\leftarrow$ the square *move* lands on

if *attacking_square* is defended **then**

return False

end if

attacked_squares $\leftarrow$ squares attacked by *attacking_square*

undefended $\leftarrow$ 0

higher_value $\leftarrow$ 0

for all *attacked_square* in *attacked_squares* **do**

if value of piece at *attacked_square* > value of piece at *attacking_square* **then**

higher_value $\leftarrow$ *higher_value* + 1

end if

if *attacked_square* is not defended **then**

undefended $\leftarrow$ *undefended* + 1

end if

end for

return *undefended* $\geq$ 2 **or** *higher_value* $\geq$ 2

Algorithm 2 Count Legal Attackers

Require: *board* and *square*

Ensure: The number of pieces which legally attack the given square

count $\leftarrow$ 0

for all legal moves in *board* **do**

if the move lands on *square* **then**

count $\leftarrow$ *count* + 1

end if

end for

return *count*

Algorithm 3 Absolute Pin TRA

Require: *prev_board*, *board*, and *move*

Ensure: True if the move results in a new Absolute Pin, False otherwise

```
if the square where move lands is defended then
    return False
end if
for all opponent pieces on board do
    if not a King or a Pawn and is pinned and was not pinned in prev_board then
        return True
    end if
end for
return False
```

Algorithm 4 Relative Pin TRA

Require: *prev_board*, *board*, and *move*

Ensure: True if the move results in a new Relative Pin, False otherwise

```
if the square where move lands is defended then
    return False
end if
ranged_pieces ← all ranged pieces of player to move
for all attacking_piece in ranged_pieces do
    for all squares attacked by attacking_piece do
        attacked_piece ← piece at the attacked square
        if attacked_piece is not a Pawn then
            (xray_square, xray_piece) ← search for an x-rayed piece
            if ((attacked_piece < xray_piece and xray_piece is not the king) and
not (attacking_piece ≥ xray_piece and xray_piece is defended) and
not (there is no Pin or Pin already existed in prev_board)) then
                return True
            end if
        end if
    end for
end for
return False
```

Algorithm 5 Double Check TRA

Require: *board*

Ensure: True if there are two or more pieces giving check, False otherwise

```
return number of pieces giving check > 2
```

Algorithm 6 Discovered Check TRA

Require: *board* and *move*

Ensure: True if the move is a Discovered Check, False otherwise

```
if board is in check then
    temp ← copy of board with an opponent pawn placed at the square where move lands
    if temp is in check then
        return True
    end if
end if
return False
```

Algorithm 7 Skewer TRA

Require: *prev_board*, *board*, and *move*

Ensure: True if the move results in a Skewer, False otherwise

```
if the square where move lands is defended then
    return False
end if
ranged_pieces  $\leftarrow$  all ranged pieces of player to move
for all attacking_piece in ranged_pieces do
    for all squares attacked by attacking_piece do
        attacked_piece  $\leftarrow$  piece at the attacked square
        if attacked_piece > attacking_piece then
            (xray_square, xray_piece)  $\leftarrow$  search for x-rayed piece
            if there is no Skewer or it already existed in prev_board then
                continue
            end if
            if attacked_piece is a King then ▷ Absolute Skewer
                if (xray_piece  $\geq$  attacking_piece or xray_square is undefended) then
                    return True
                end if
            else if attacking_piece < xray_piece  $\leq$  attacked_piece then ▷ Relative Skewer
                return True
            end if
        end if
    end if
end for
return False
```

Algorithm 8 King Defence

Require: *board*, *attacking_square*, and *xray_square*

Ensure: True if the King can defend the x-rayed square by moving to a legal square, False otherwise

```
b  $\leftarrow$  copy of board
for all legal moves in board do
    if the move originates from the current turn's king then
        apply the move to b
        temp  $\leftarrow$  copy of b without the attacking piece
        if xray_square is not attacked in temp then
            return True
        end if
        undo the move from b
    end if
end for
return False
```

Algorithm 9 Undermine TRA

Require: *board* and *moves*

Ensure: True if the sequence of moves is an Undermine tactic, False otherwise

```
m1, m2, m3  $\leftarrow$  first three moves from moves
if m1 and m2 land on the same square and m1 is capture and m3 is capture then
    defended_squares  $\leftarrow$  squares defended by the piece that was captured in m1
    if m3 in defended_squares and m3 and m1 land on different squares then
        return True
    end if
end if
return False
```

Algorithm 10 Get X-rayed Piece

Require: *board*, *attacking_square*, and *attacked_square*
Ensure: Tuple containing the square and piece of an x-rayed piece, or (-1, Pawn) if none
 ray $\leftarrow$ list of squares in the ray between *attacking_square* and *attacked_square*
 if *attacking_square* < *attacked_square* **then**
 xray $\leftarrow$ squares after *attacked_square* in *ray*
 else
 xray $\leftarrow$ squares before *attacked_square* in *ray*
 reverse *xray*
 end if
 for all squares in *xray* **do**
 if there is a piece at the square **then**
 return square and piece at square **if** opponent piece **else** *invalid*
 end if
 end for
 return *invalid*

Algorithm 11 is X-ray

Require: *board*, *attacking_square*, *attacked_square*, and *xray_square*
Ensure: True if the three pieces are on the same ray without any wedging pieces, False otherwise
 ray $\leftarrow$ squares in the ray between *attacking_square* and *attacked_square*
 if *attacking_square*, *attacked_square*, and *xray_square* are all in *ray* **then**
 between $\leftarrow$ squares between *attacking_square* and *attacked_square*
 if *between* is empty **or** no pieces placed on *between* **then**
 return True
 end if
 end if
 return False

Algorithm 12 Tactic Extraction

1: **Input:** DataFrame *df* containing chess game data
2: **Output:** DataFrame *tactics_df* containing extracted tactics
3: **for** each game in *df* **do**
4: Extract move timestamps and calculate move times
5: **for** each move in the game **do**
6: **if** move is in half-move range **then**
7: Run a Stockfish analysis on the previous position
8: Search for tactic in the engine line, store if found
9: **if** Player's move was not the top engine line **then**
10: Search for tactic in player's move
11: **if** tactic found and move is 'good' **then**
12: Store tactic information
13: **end if**
14: **end if**
15: **end if**
16: **end for**
17: **end for**

Algorithm 13 Move Classification

```
1: function ISWINNINGMOVE(mainline, nextline)
2:   if (mainline.score > 1000 and nextline.score > 1000) or
3:     (mainline.score < -300 and nextline.score < -300) then
4:     return False
5:   end if
6:   return (mainline.score - nextline.score) > 300
7: end function
```
